

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

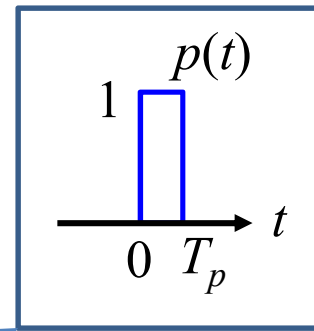
Sampling and Analog-to-Digital Conversion

Practical Signal Reconstruction (Interpolation)

$$\tilde{G}(f) = P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$

$$G(f) = E(f) \tilde{G}(f)$$

$$= E(f) P(f) \frac{1}{T_s} \sum_n G(f - nf_s)$$



$$p(t) = \Pi\left(\frac{t - 0.5T_p}{T_p}\right)$$

$$P(f) = T_p \text{sinc}(\pi f T_p) e^{-j\pi f T_p}$$

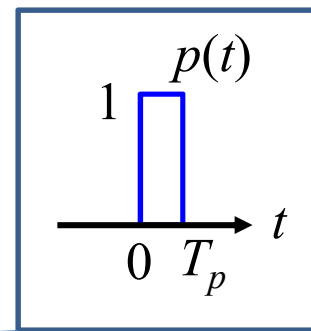
$$E(f) = \begin{cases} T_s / P(f) & |f| \leq B \\ \text{Flexible} & B < |f| \leq f_s - B \\ 0 & |f| \geq f_s - B \end{cases}$$

Practical Signal Reconstruction (Interpolation)

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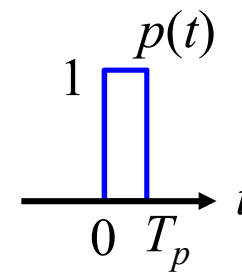
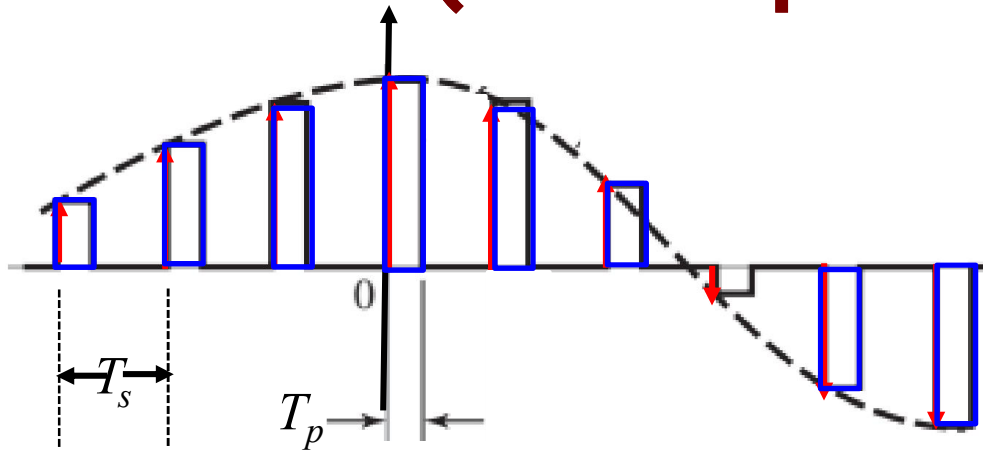
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$$\longrightarrow |E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

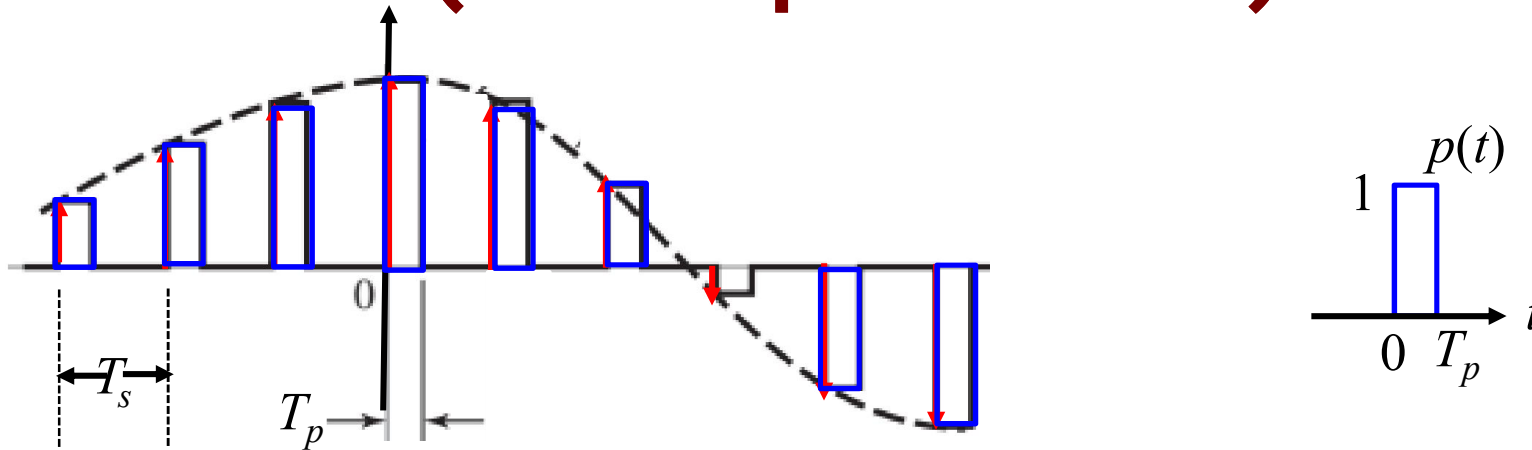
Practical Signal Reconstruction (Interpolation)



$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

T_p must be $< 1/B$,
otherwise signal will be
distorted

Practical Signal Reconstruction (Interpolation)

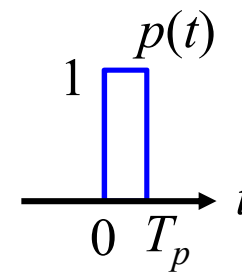
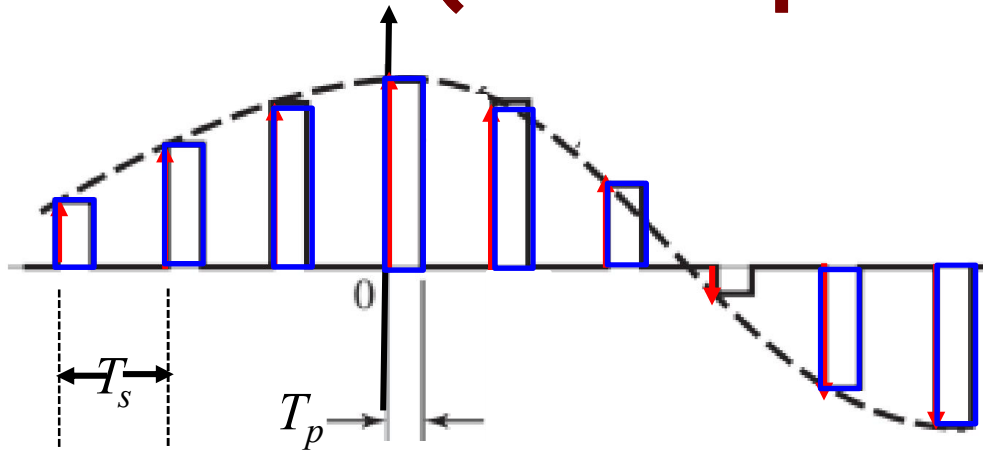


$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \quad |f| \leq B$$

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distorted

$T_p > 0$, otherwise $E(f)$ will
have infinite gain

Practical Signal Reconstruction (Interpolation)

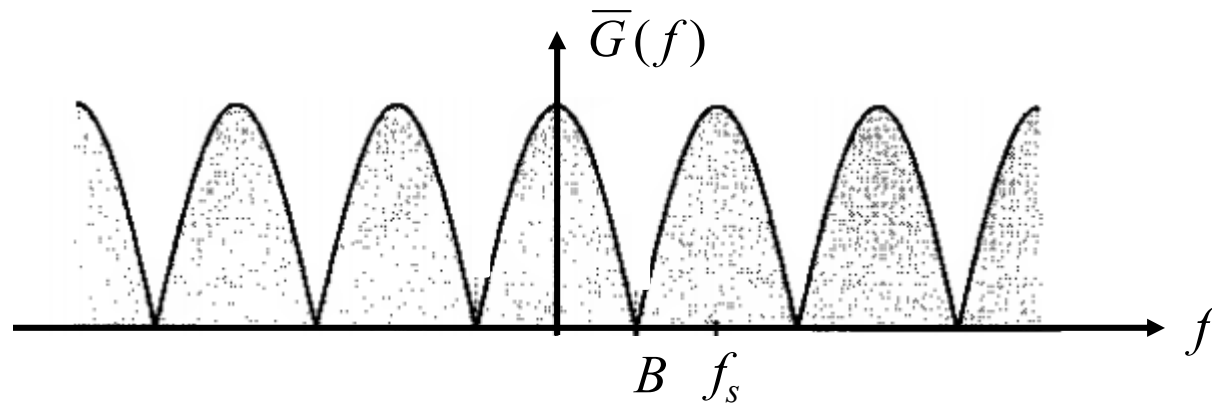


in special case when T_p is very small,

$$|E(f)| = T_s \cdot \left| \frac{\pi f}{\sin(\pi f T_p)} \right| \approx \frac{T_s}{T_p} \quad |f| \leq B$$

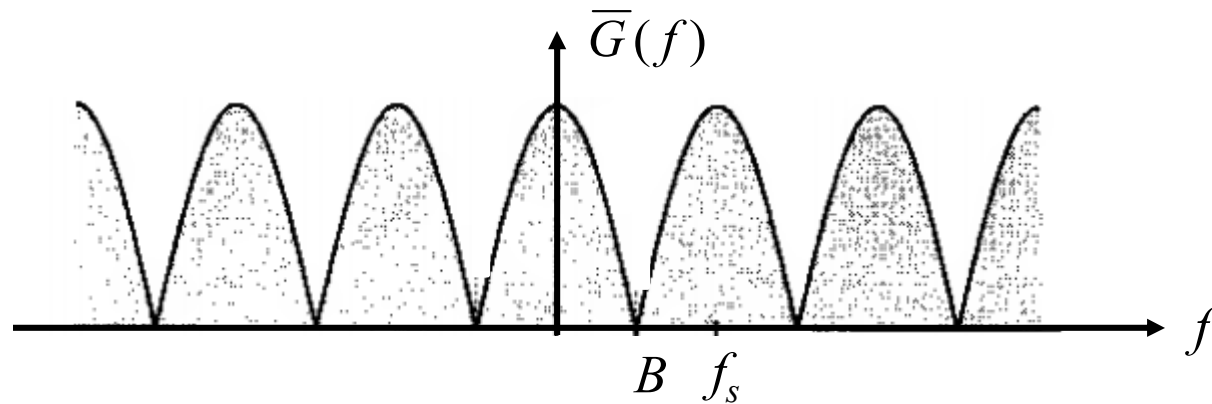
Realizability of Reconstruction Filters

Spectrum $\overline{G}(f)$ when signal $g(t)$ is
sampled at the Nyquist rate $f_s = 2B$



Realizability of Reconstruction Filters

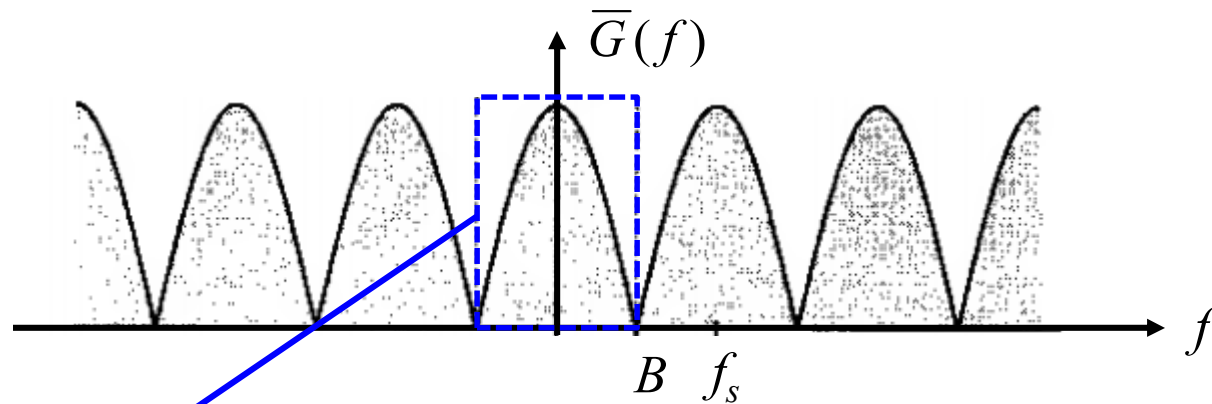
Spectrum $\overline{G}(f)$ when signal $g(t)$ is sampled at the Nyquist rate $f_s = 2B$



Repetitions of $G(f)$ **without any gap** between successive cycles

Realizability of Reconstruction Filters

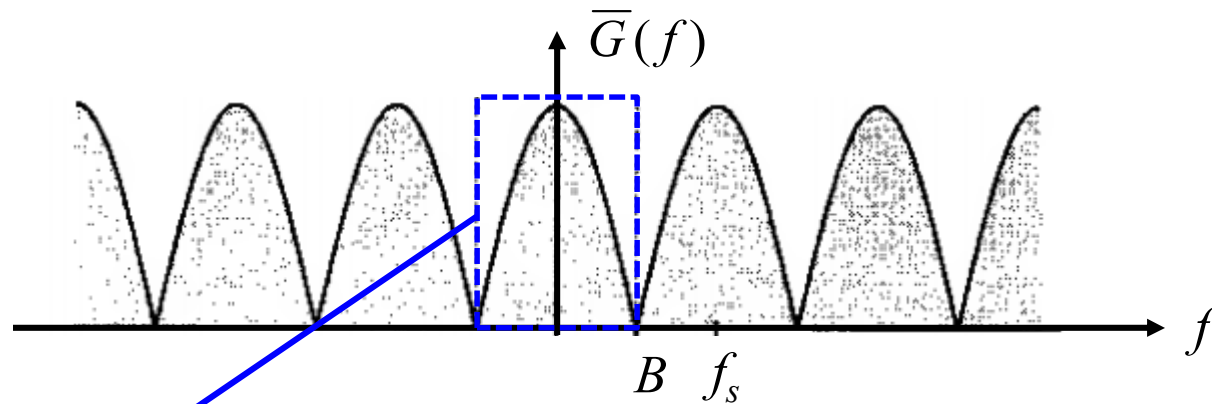
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Ideal reconstruction
filter (*Box filter*)

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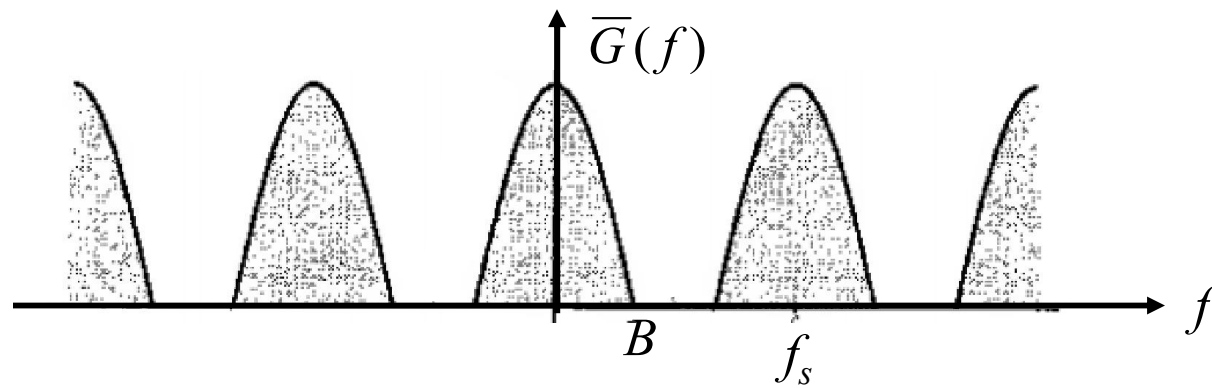


Ideal reconstruction
filter (*Box filter*)

But box filter is unrealizable; can be
approximated only with infinite time delay

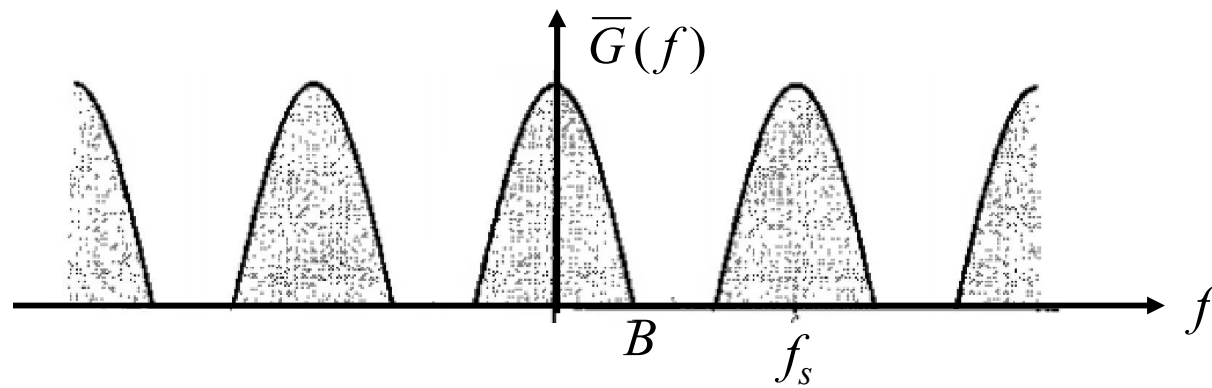
Realizability of Reconstruction Filters

Alternate: Sample at a rate higher than
Nyquist rate $f_s > 2B$



Realizability of Reconstruction Filters

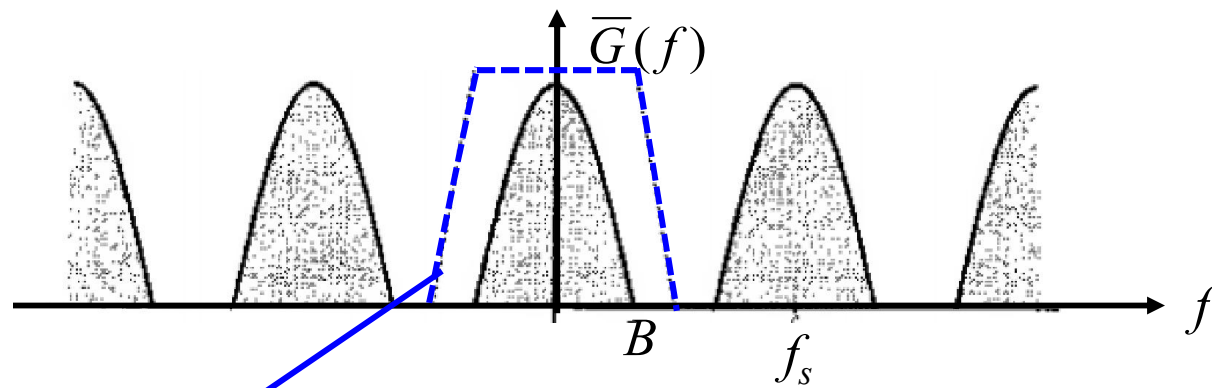
Alternate: Sample at a rate higher than
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Repetitions of $G(f)$ with finite gap
between successive cycles

Realizability of Reconstruction Filters

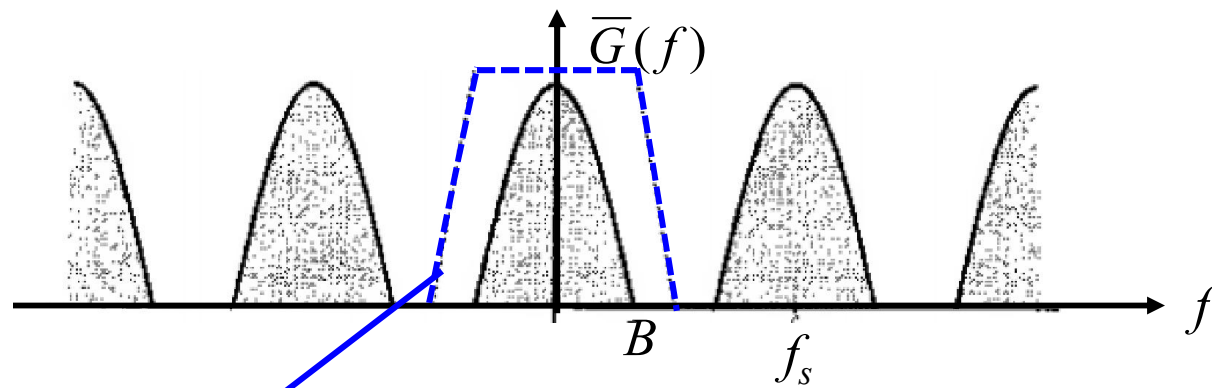
Alternate: Sample at a rate higher than Nyquist rate $f_s > 2B$



Low pass filter with gradual cutoff characteristics

Realizability of Reconstruction Filters

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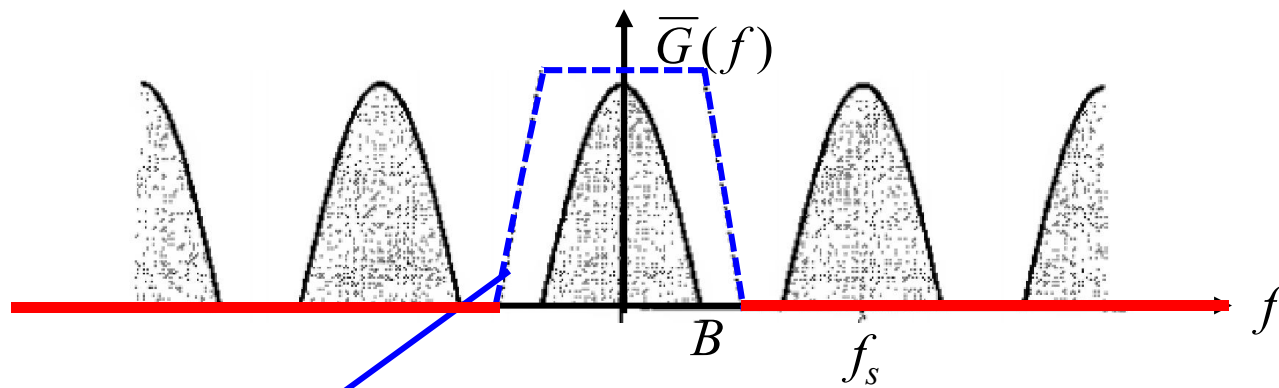


Low pass filter with gradual cutoff characteristics

Advantage: requires smaller time delay

Realizability of Reconstruction Filters

Alternate: Sample at a rate higher than Nyquist rate $f_s > 2B$



Low pass filter with gradual cutoff characteristics

Disadvantage: filter gain is zero beyond first cycle of $G(f)$ $\rightarrow$ fails to satisfy Paley-Wiener criteria

Realizability of Reconstruction Filters

Findings:

- Exact recovery of $g(t)$ from samples is impossible, even $f_s > 2B$
- As f_s increases, recovered signal approaches $g(t)$

Wrong Assumption on Bandwidth: Treachery of Aliasing

Definition of aliasing

- A **false** or **assumed** identity: same person, different names
- an **alternative name** or **label** that refers to a file, command, address, or other item, and can be used to locate or access it.

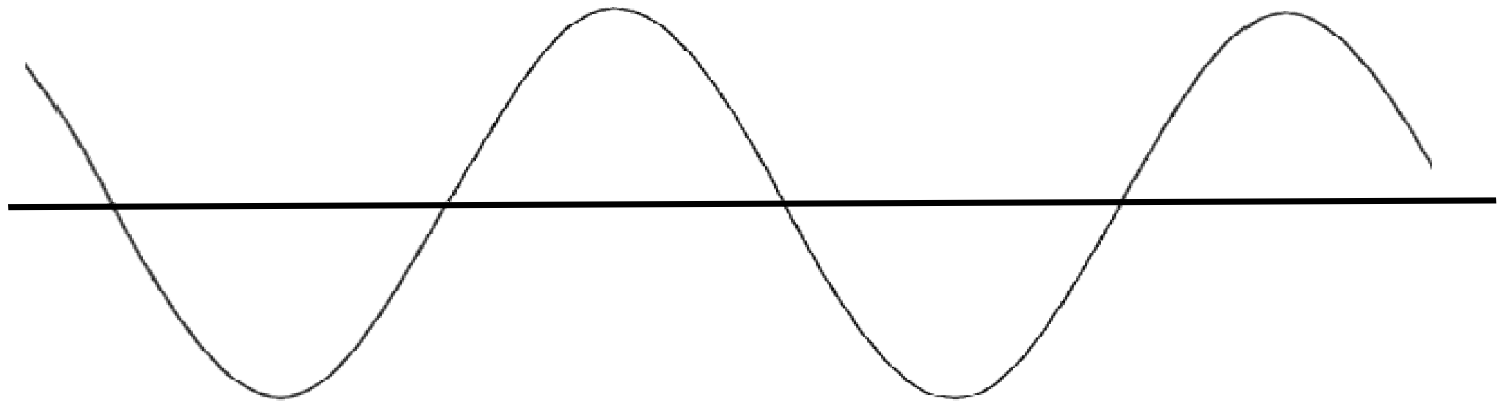
Wrong Assumption on Bandwidth: Treachery of Aliasing

Definition of aliasing in signal processing

- each of a group of signal frequencies that, when sampled at a given uniform rate, would give the same set of sampled values, and thus might be incorrectly substituted for one another when reconstructing the original signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

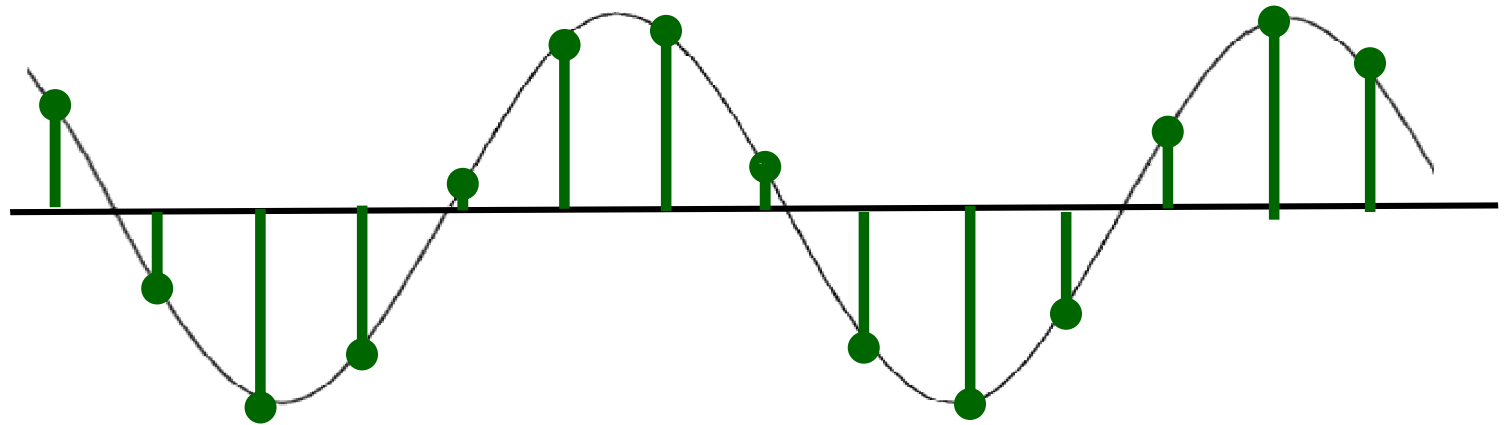
Definition of aliasing



A signal of 60 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

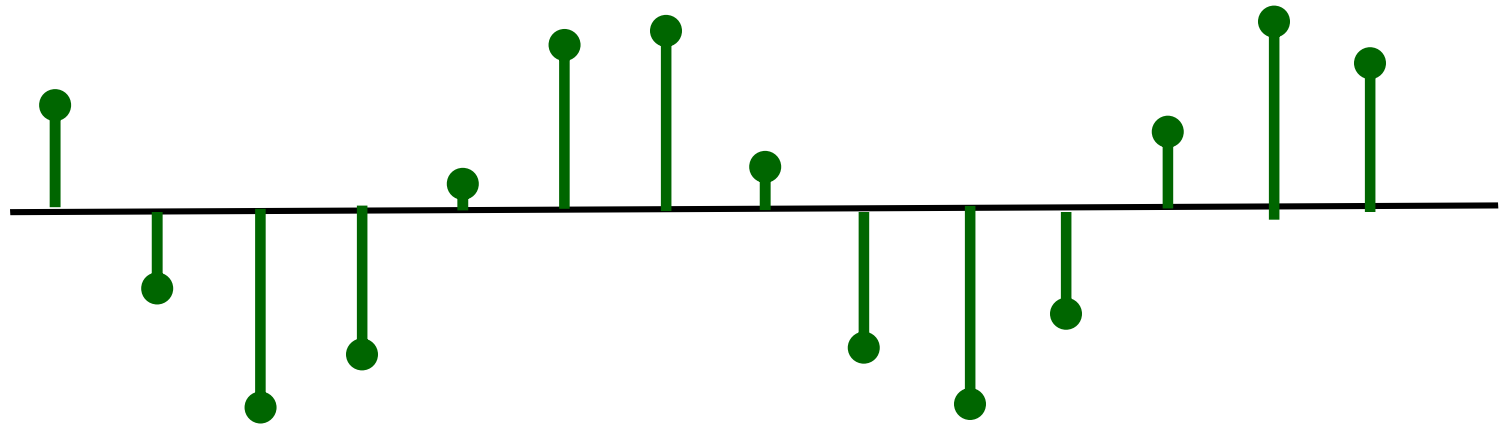
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

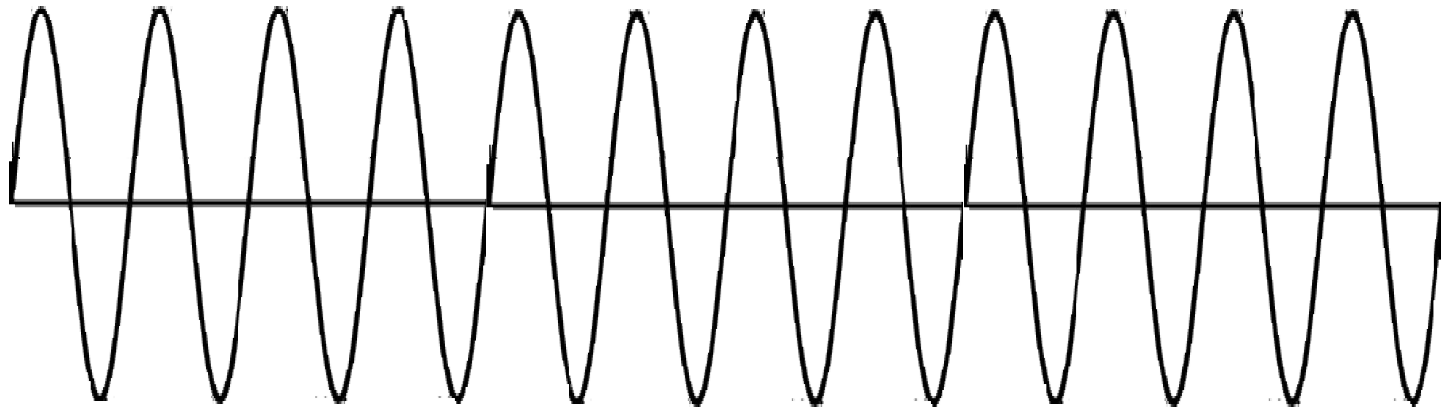
Definition of aliasing



Sampled Signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

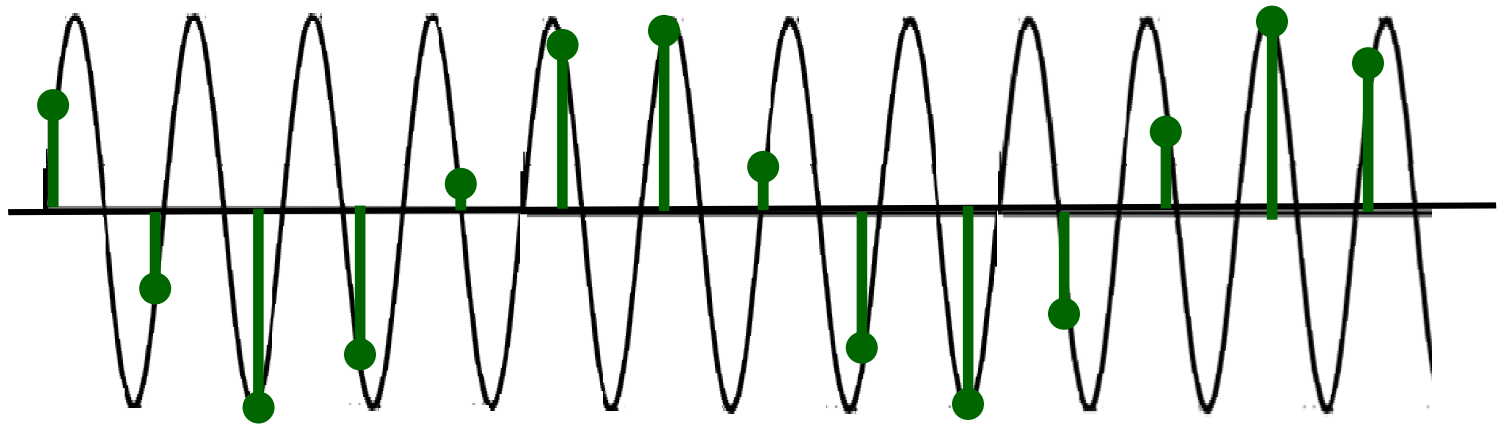
Definition of aliasing



Another signal with 340 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

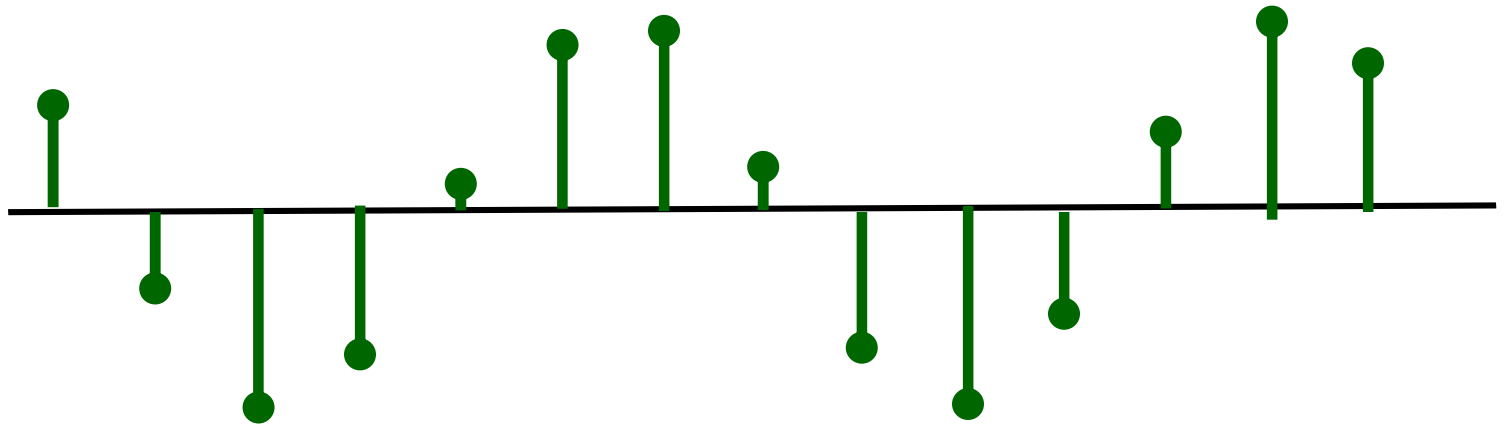
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

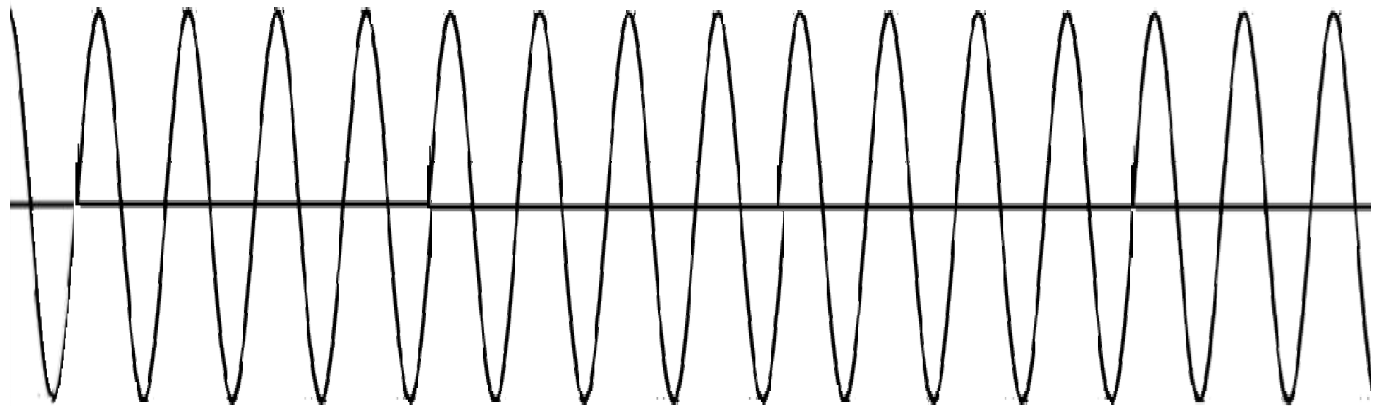
Definition of aliasing



We got the same sampled signal

Wrong Assumption on Bandwidth: Treachery of Aliasing

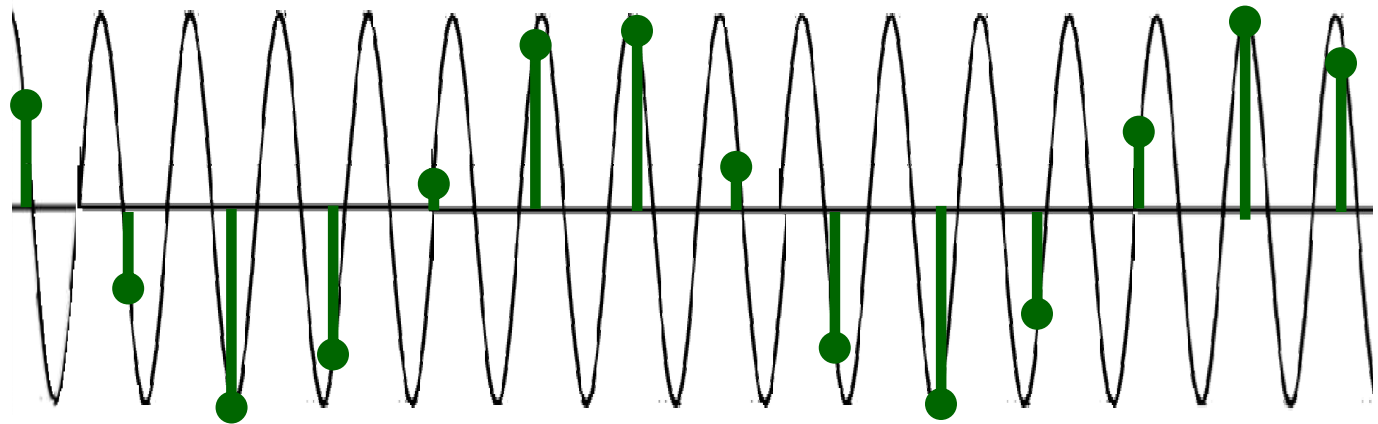
Definition of aliasing



The third signal: 460 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

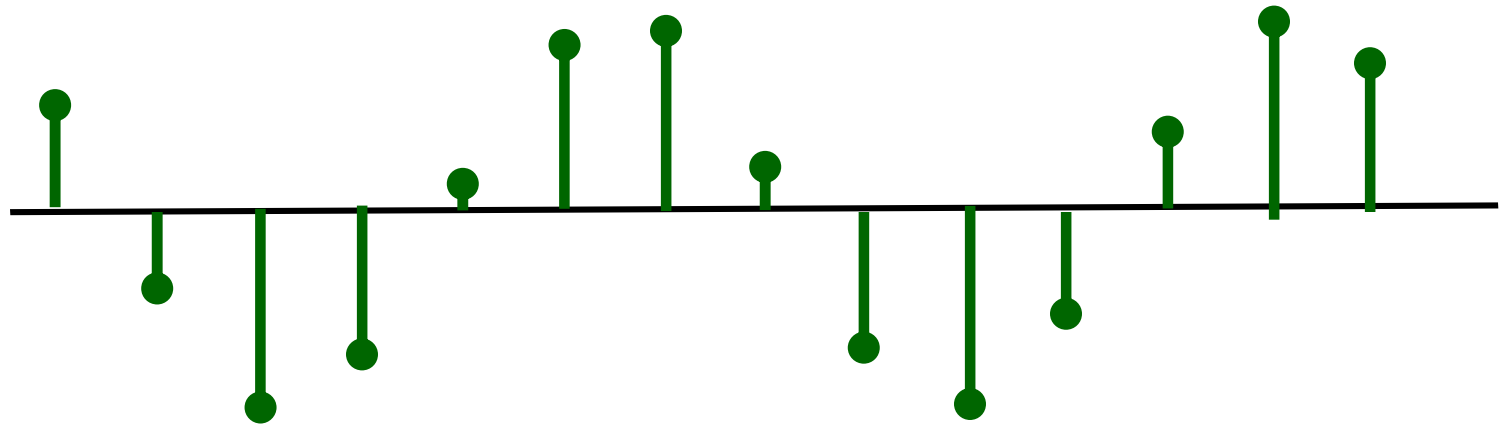
Definition of aliasing



Sampled at 400 Hz frequency

Wrong Assumption on Bandwidth: Treachery of Aliasing

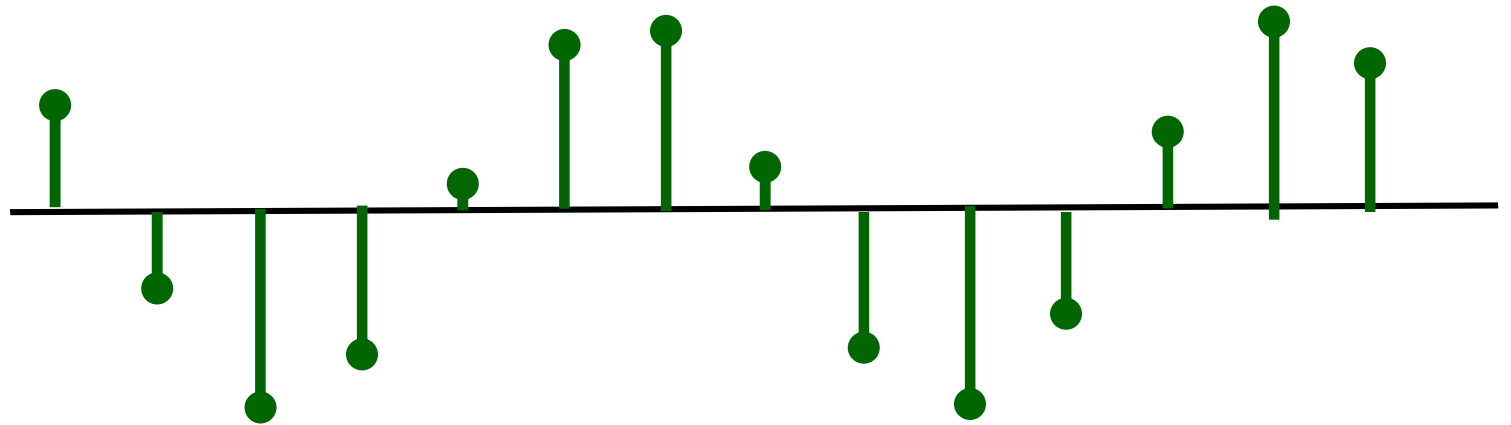
Definition of aliasing



No difference: the same sampled signal is obtained

Wrong Assumption on Bandwidth: Treachery of Aliasing

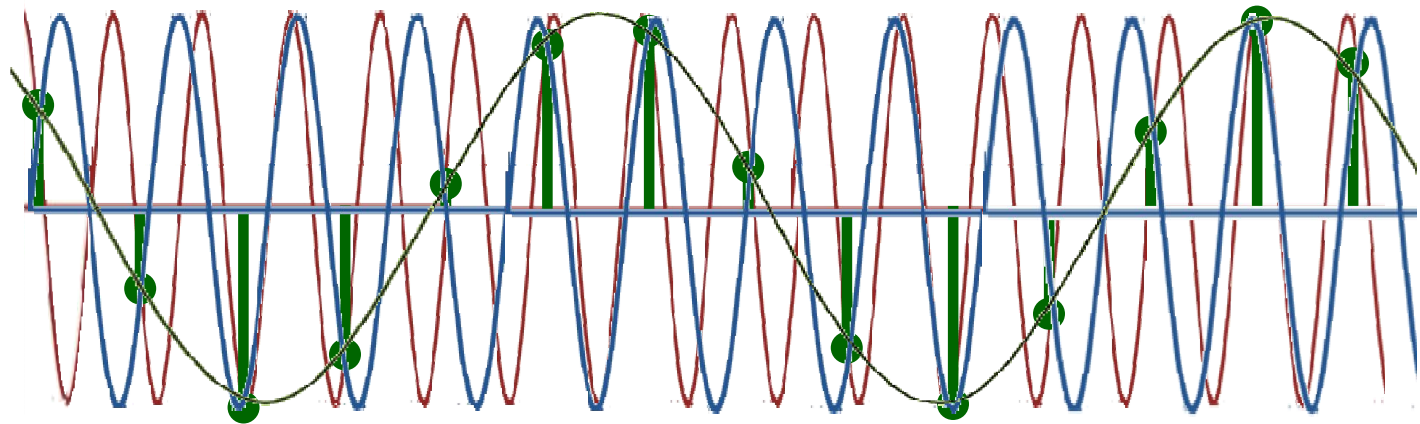
Definition of aliasing



While reconstructing, these samples may generate all three signals

Wrong Assumption on Bandwidth: Treachery of Aliasing

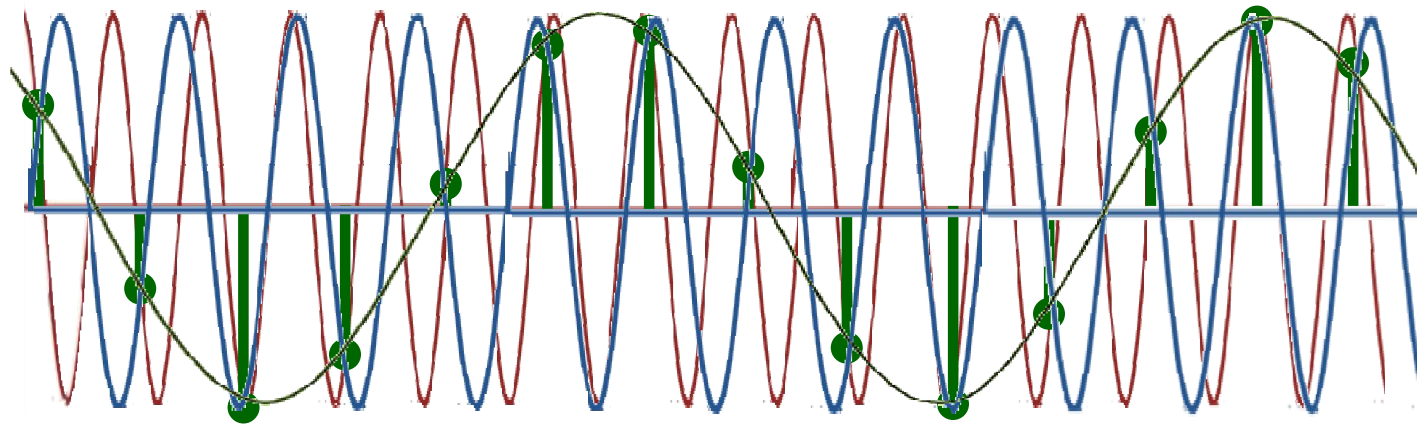
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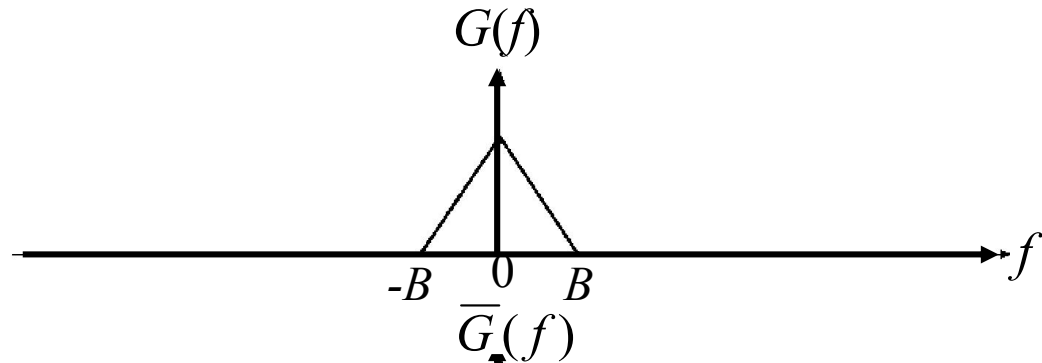
60, 340, and 460, are aliased frequencies for 400 Hz
sampling rate

Wrong Assumption on Bandwidth: Treachery of Aliasing

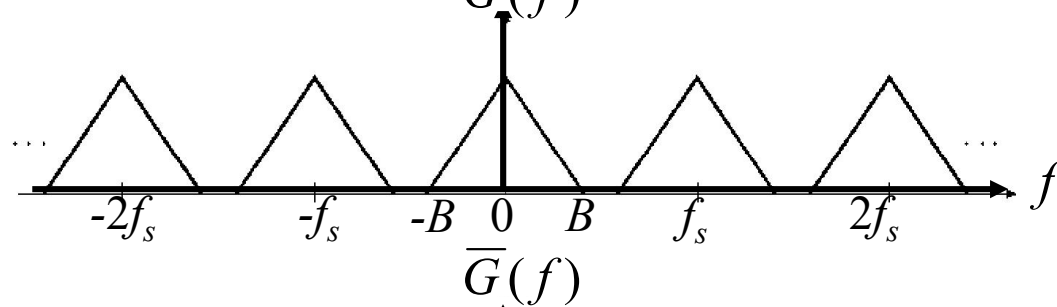
Definition of aliasing

- Aliasing occurs when sampling rate is $<$ Nyquist rate
- Spectral cycles overlaps

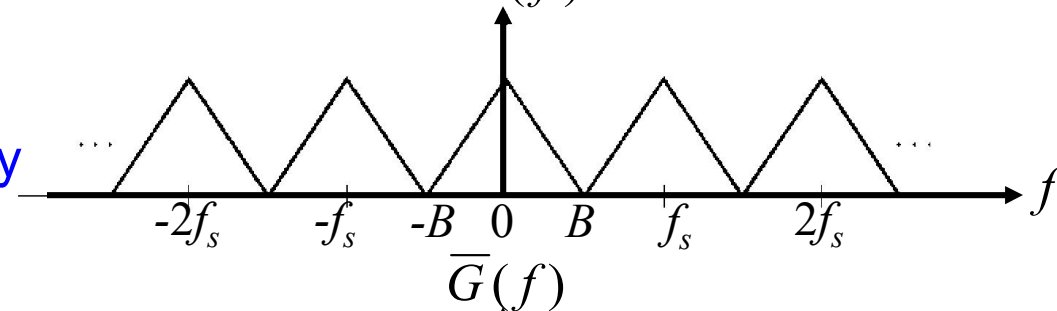
Spectrum of
unsampled $g(t)$



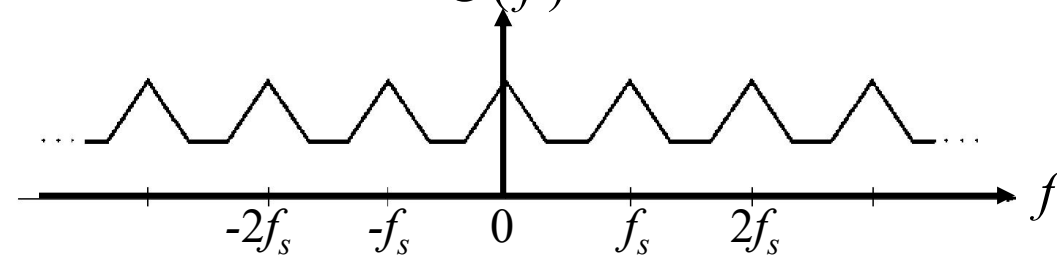
Spectrum of over
sampled $g(t)$



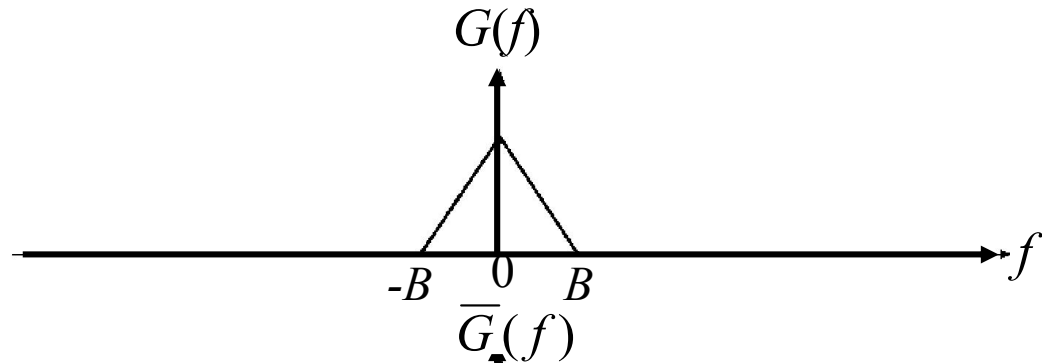
Spectrum of critically
sampled $g(t)$



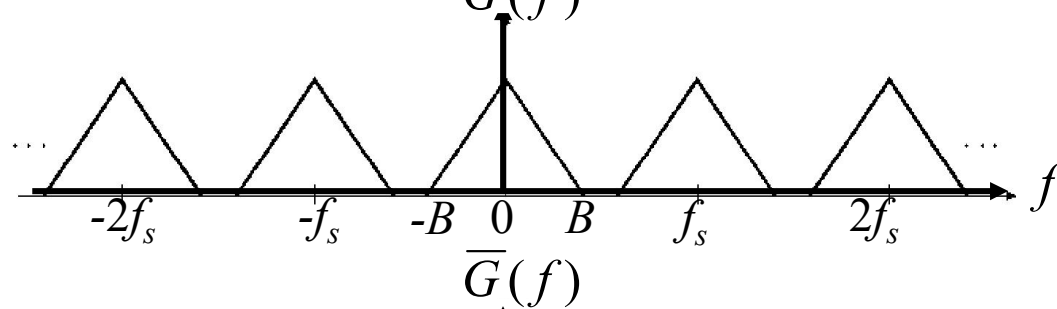
Spectrum of under
sampled $g(t)$



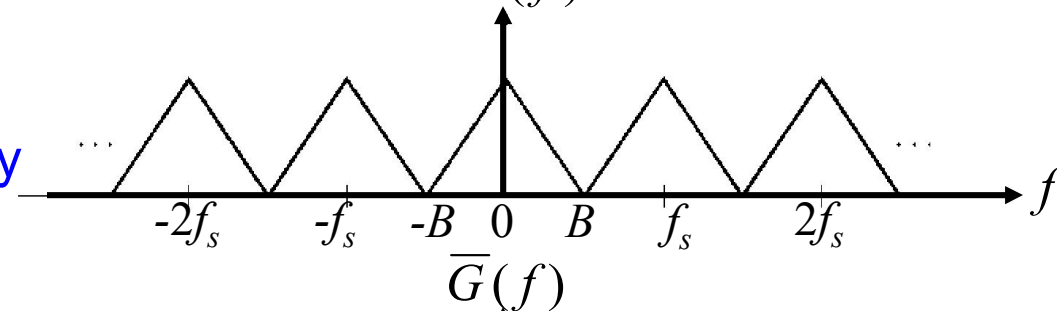
Spectrum of
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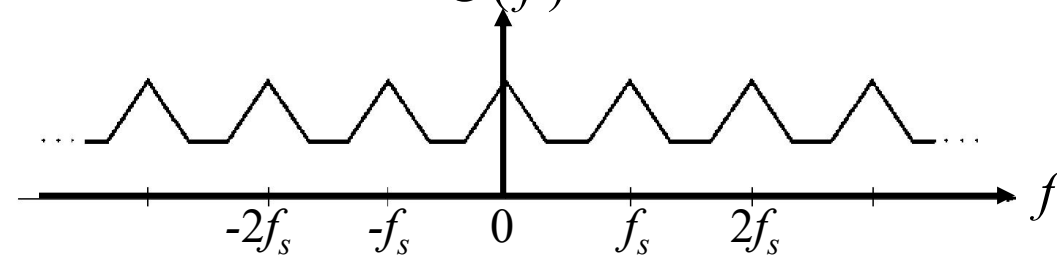
Spectrum of over
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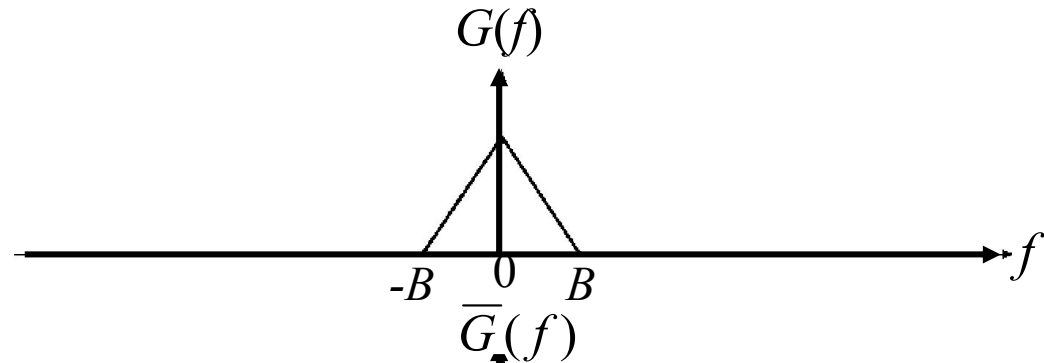
Spectrum of critically
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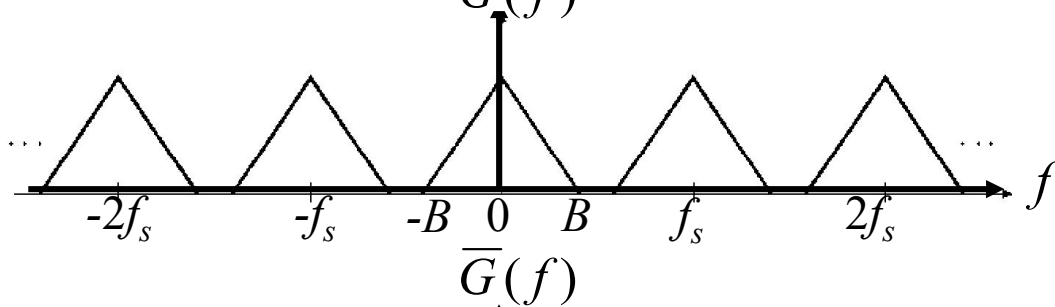
Overlapping cycles
of $G(f)$



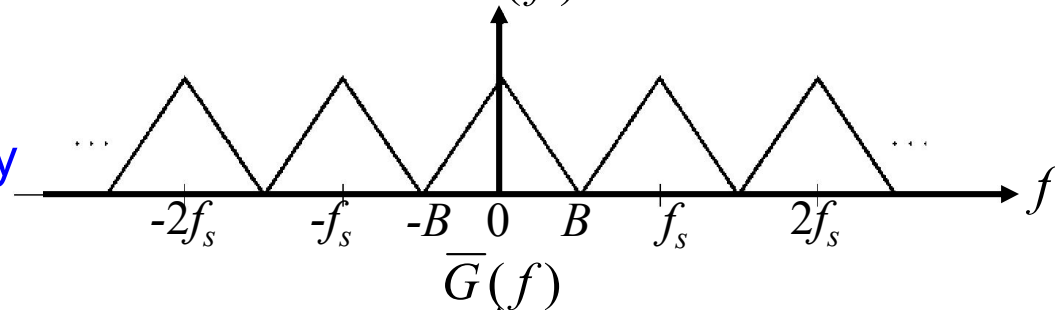
Spectrum of
unsampled $g(t)$



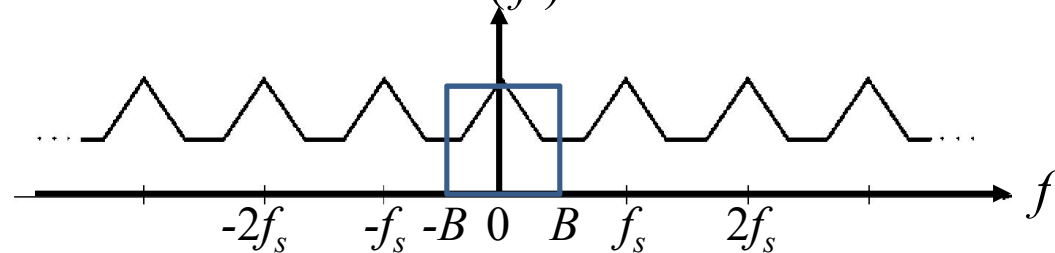
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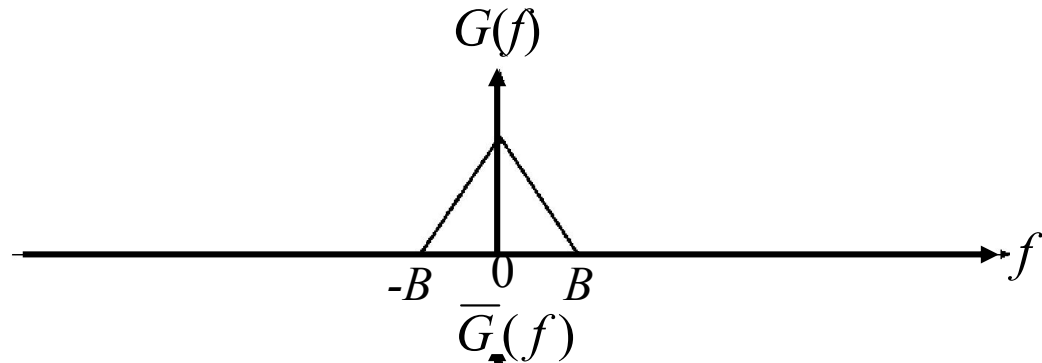
Spectrum of critically
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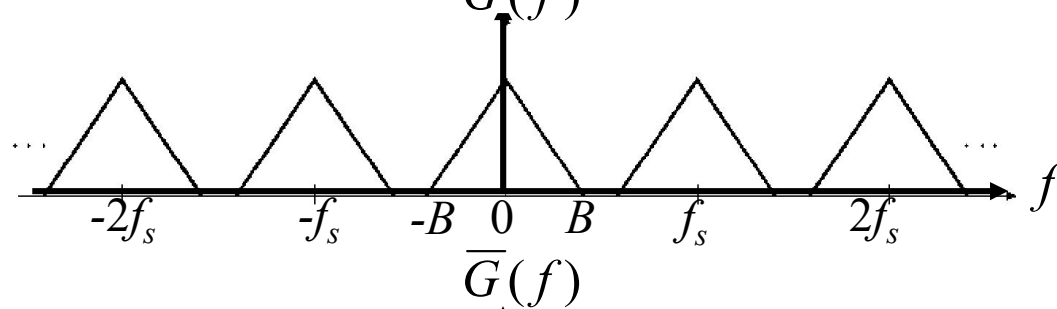
a complete cycle of $G(f)$
cannot be recovered



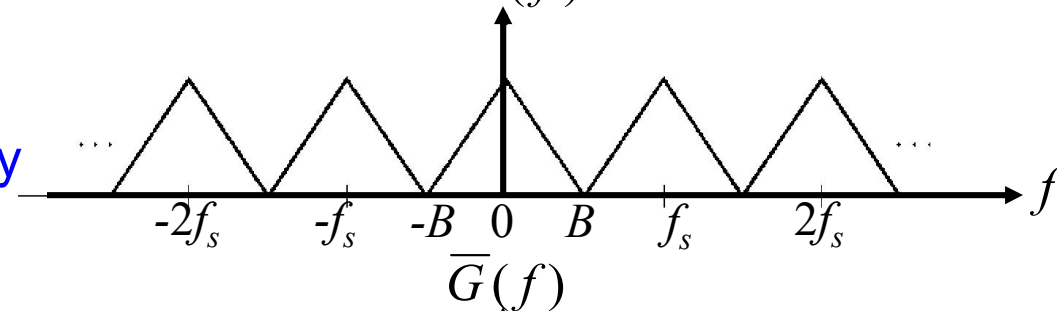
Spectrum of
unsampled $g(t)$



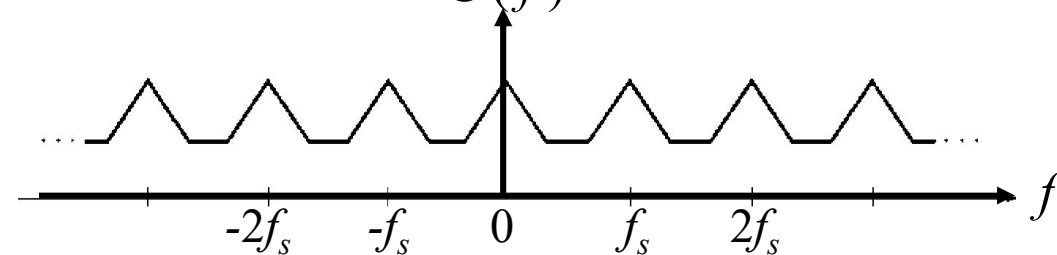
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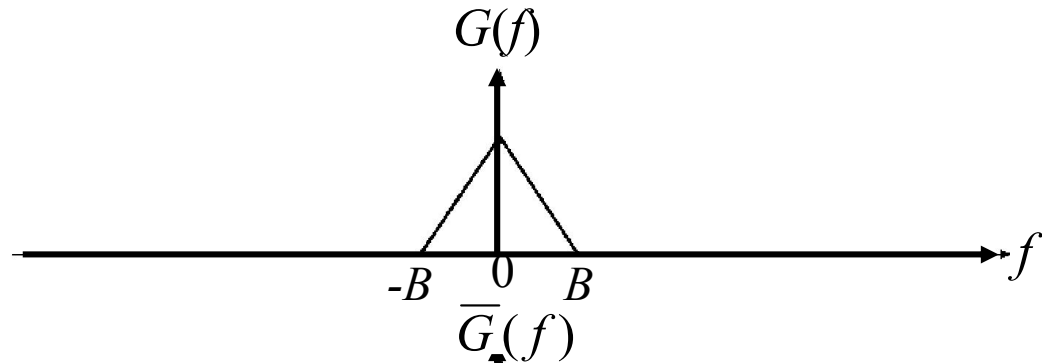
Spectrum of critically
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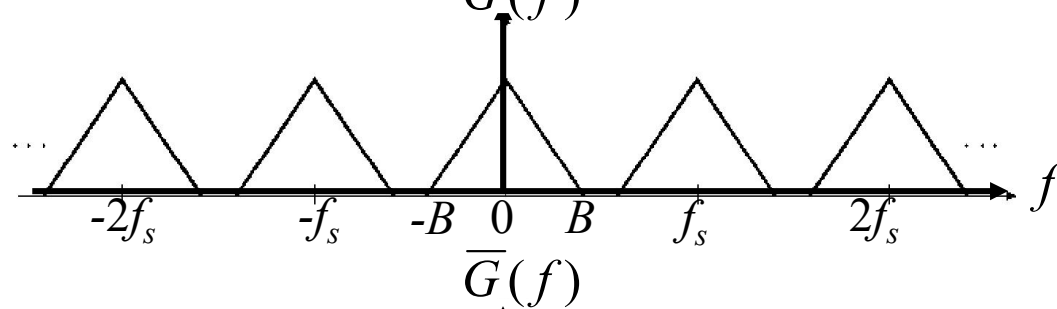
Original $g(t)$ may not be
recovered, too



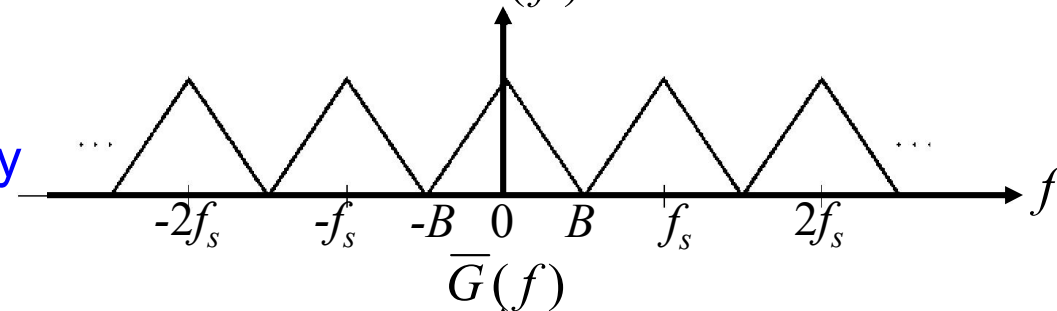
Spectrum of
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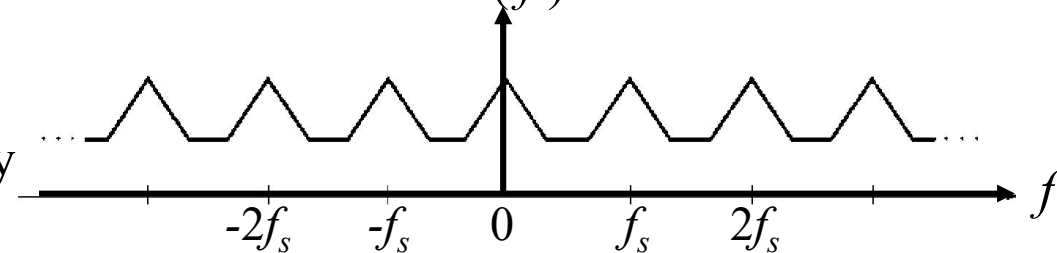
Spectrum of over
sampled $g(t)$



Spectrum of critically
sampled $g(t)$



Rather aliased signals may
be reconstructed



Wrong Assumption on Bandwidth: Treachery of Aliasing

While sampling and reconstructing,

- Signal $g(t)$ is assumed to be band-limited to B

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However,

- all practical signals are time-limited

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However,

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- time limited signal CANNOT be band-limited

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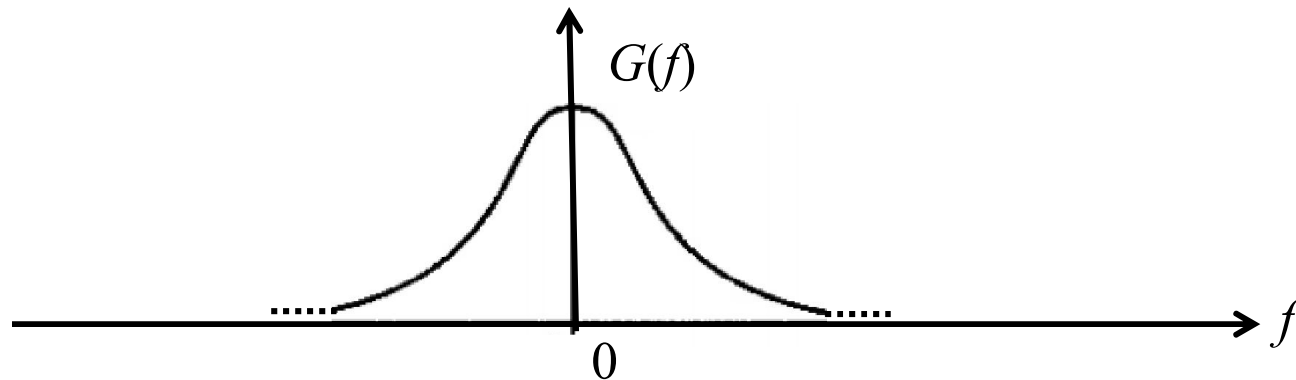
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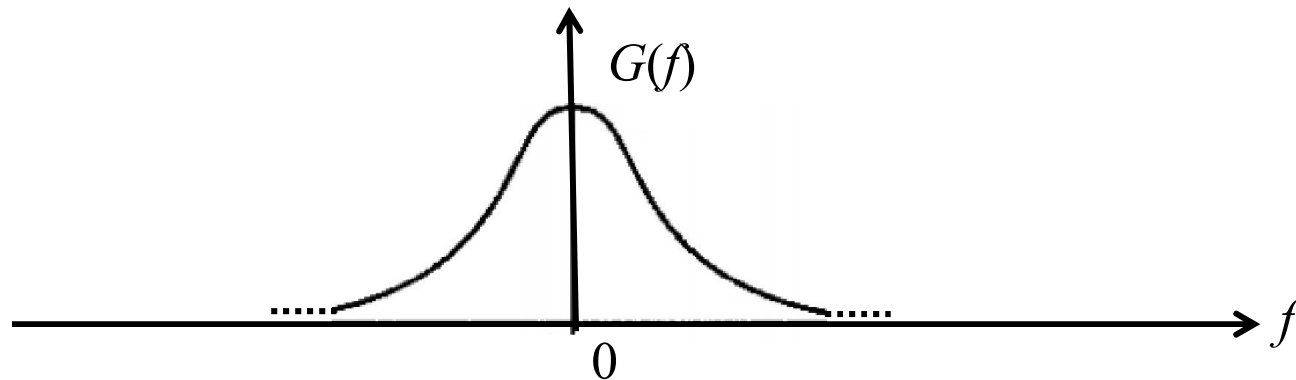
- all practical signals are time-limited
- time limited signal CANNOT be band-limited
- practical signals are therefore NON-band-limited

Wrong Assumption on Bandwidth: Treachery of Aliasing



A practical signal with
infinite bandwidth (B)

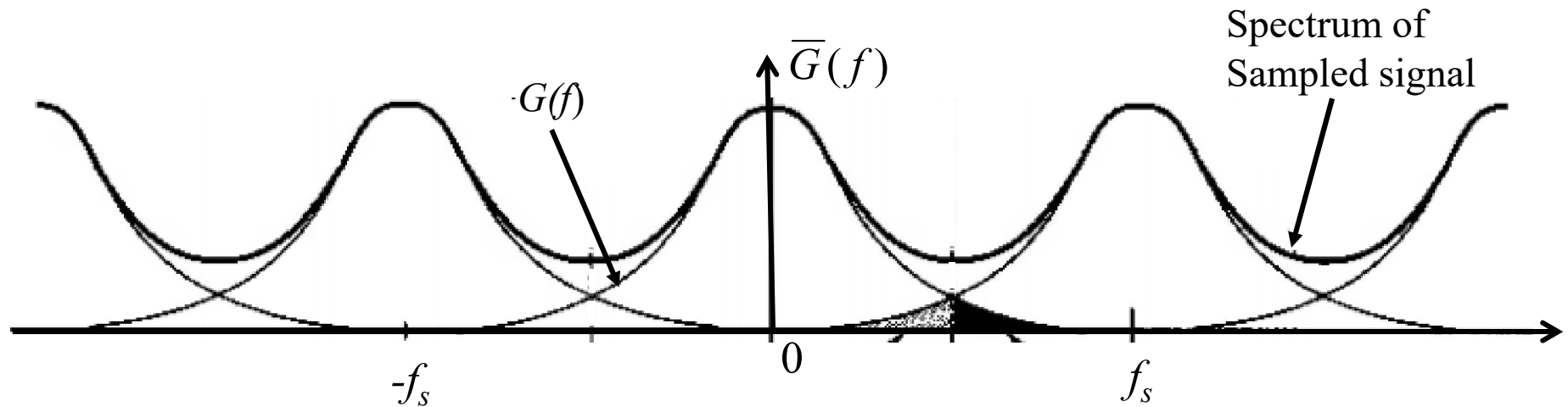
Wrong Assumption on Bandwidth: Treachery of Aliasing



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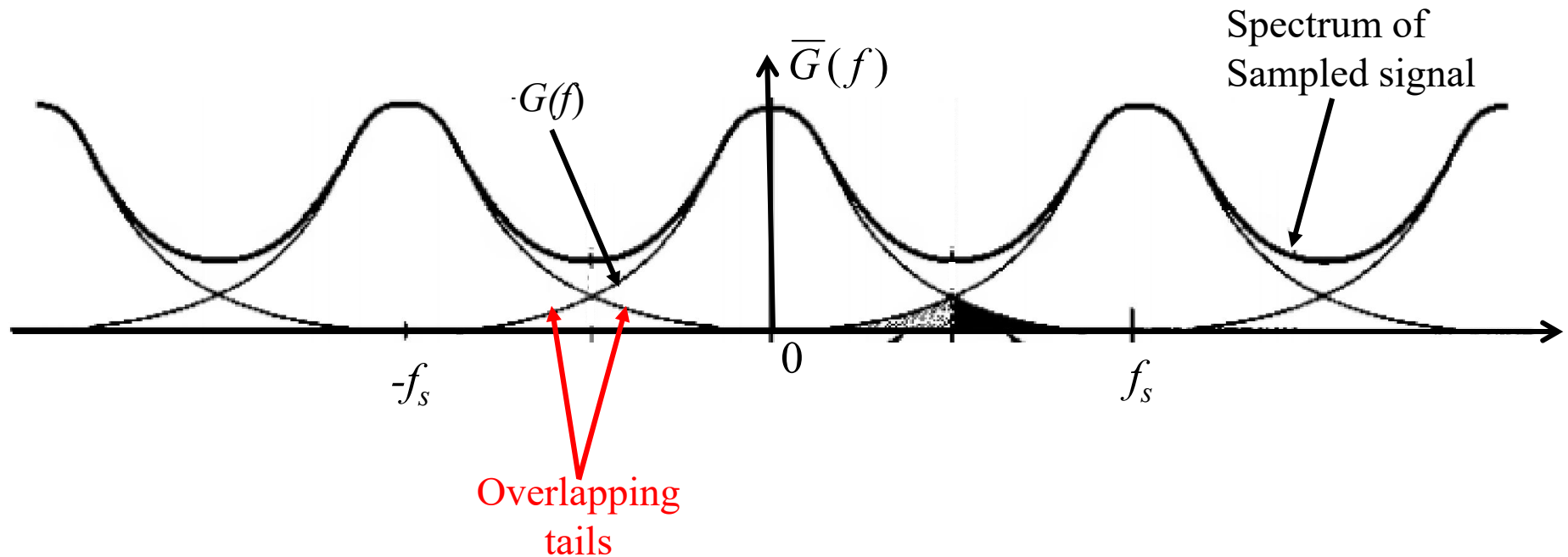
Sampling will result $\bar{G}(f)$ consisting of overlapping $G(f)$ at every f_s

Wrong Assumption on Bandwidth: Treachery of Aliasing



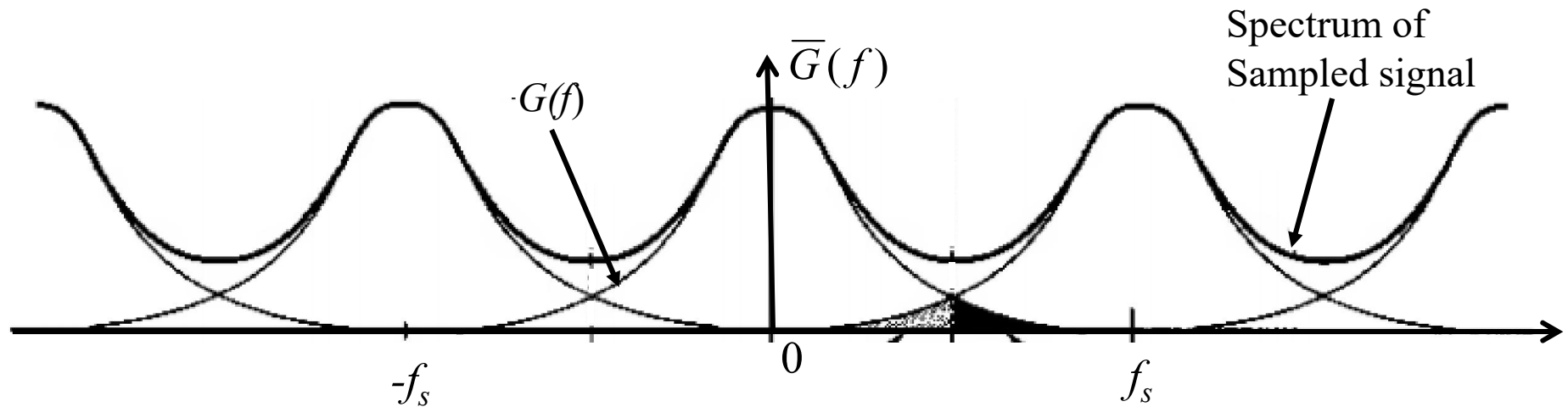
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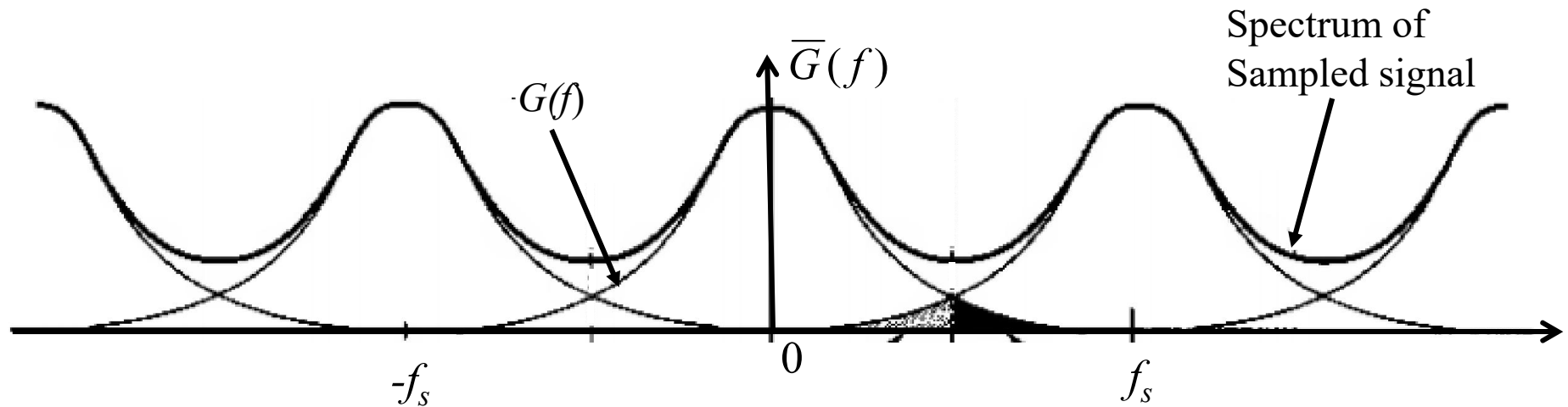
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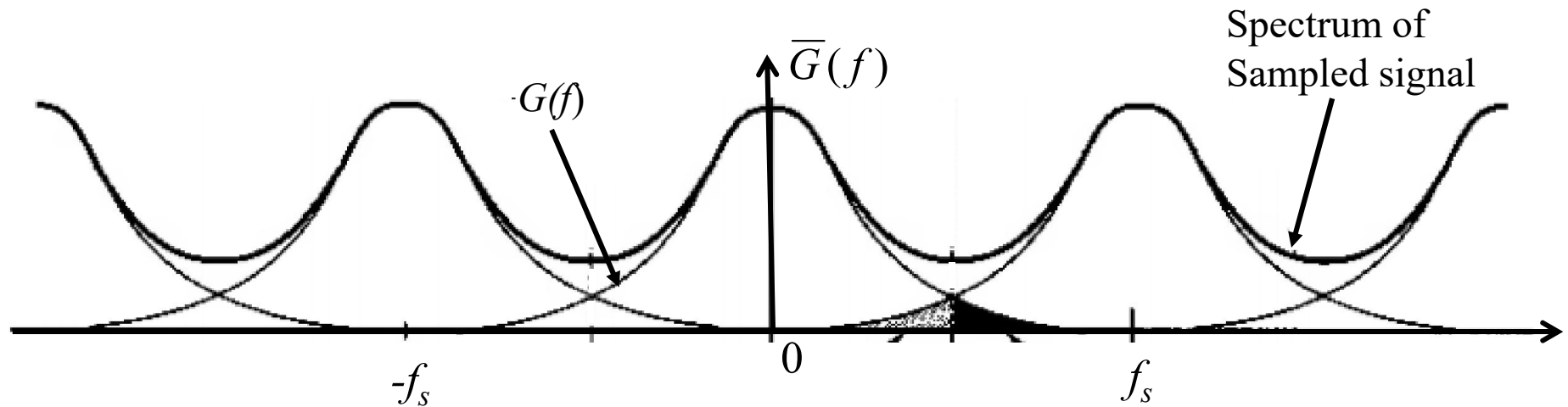
Spectral **overlap is unavoidable** as bandwidth is infinite

Wrong Assumption on Bandwidth: Treachery of Aliasing



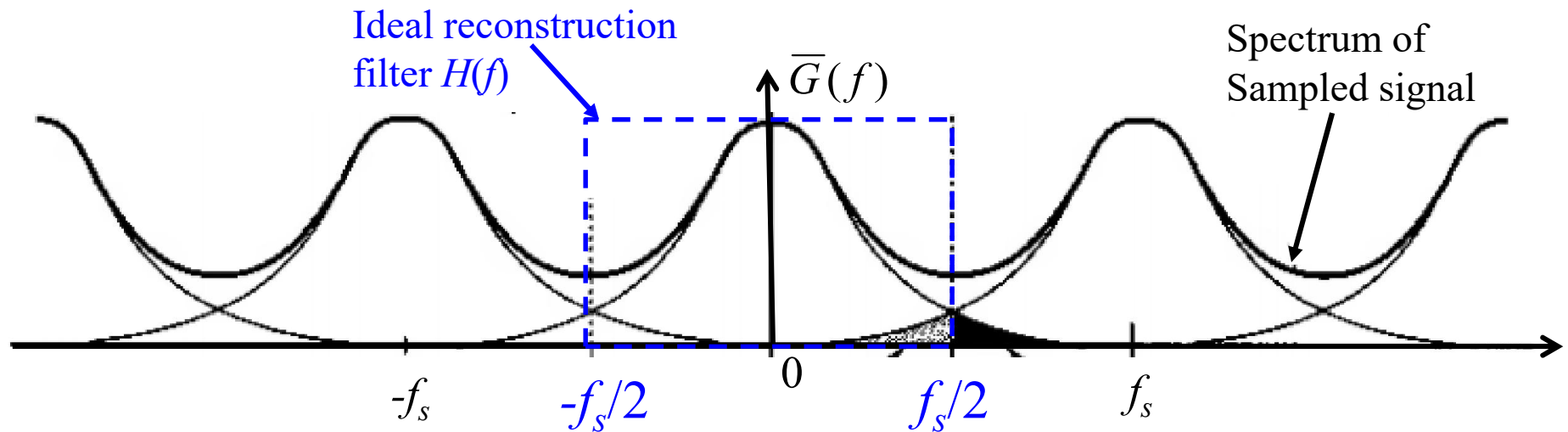
Higher sampling rate **ONLY** reduces overlap; but **cannot eliminate**

Wrong Assumption on Bandwidth: Treachery of Aliasing



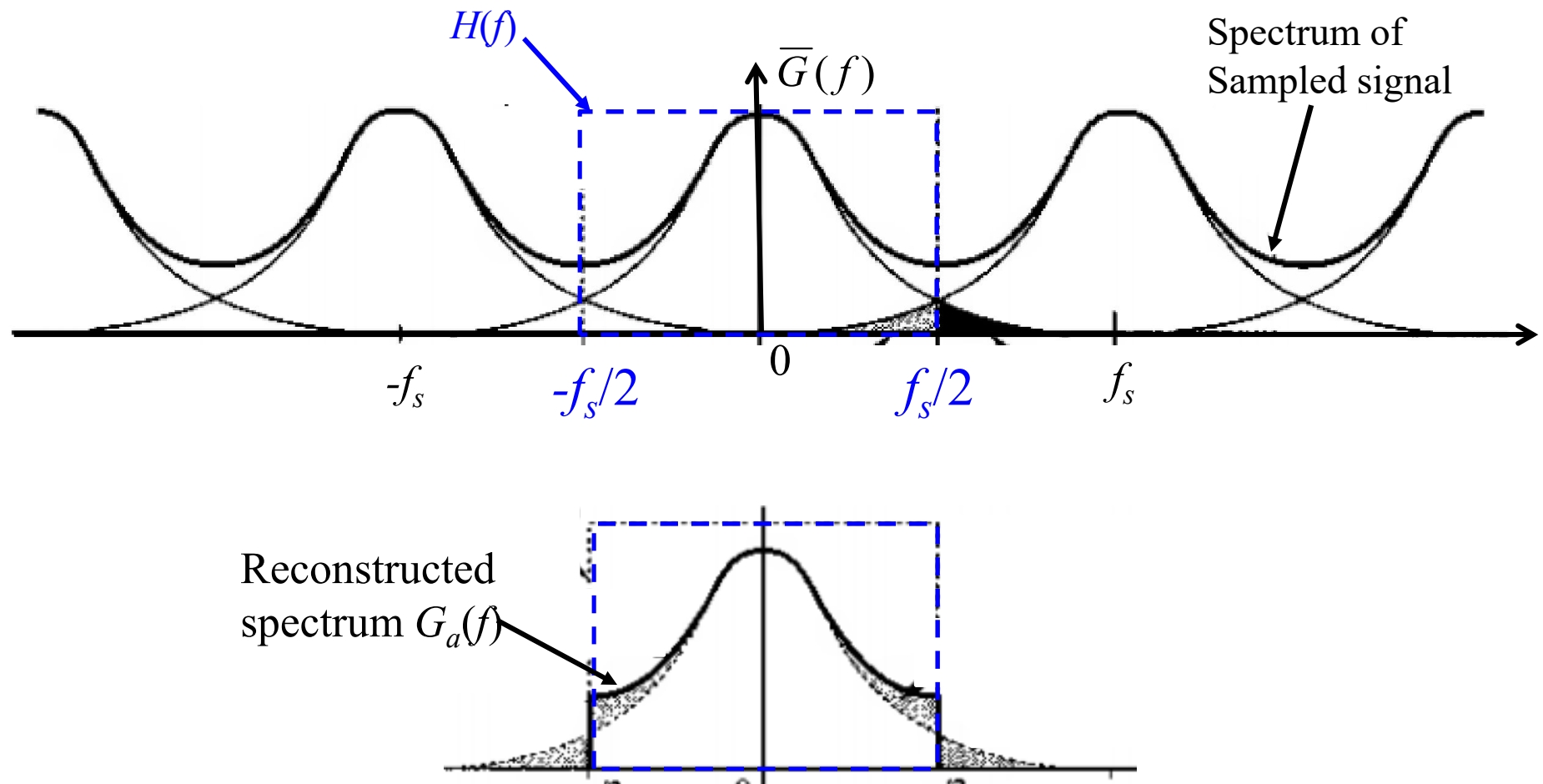
$\bar{G}(f)$ has NO complete information of $G(f)$
 $g(t)$ cannot be fully recovered from $\bar{g}(t)$

Wrong Assumption on Bandwidth: Treachery of Aliasing

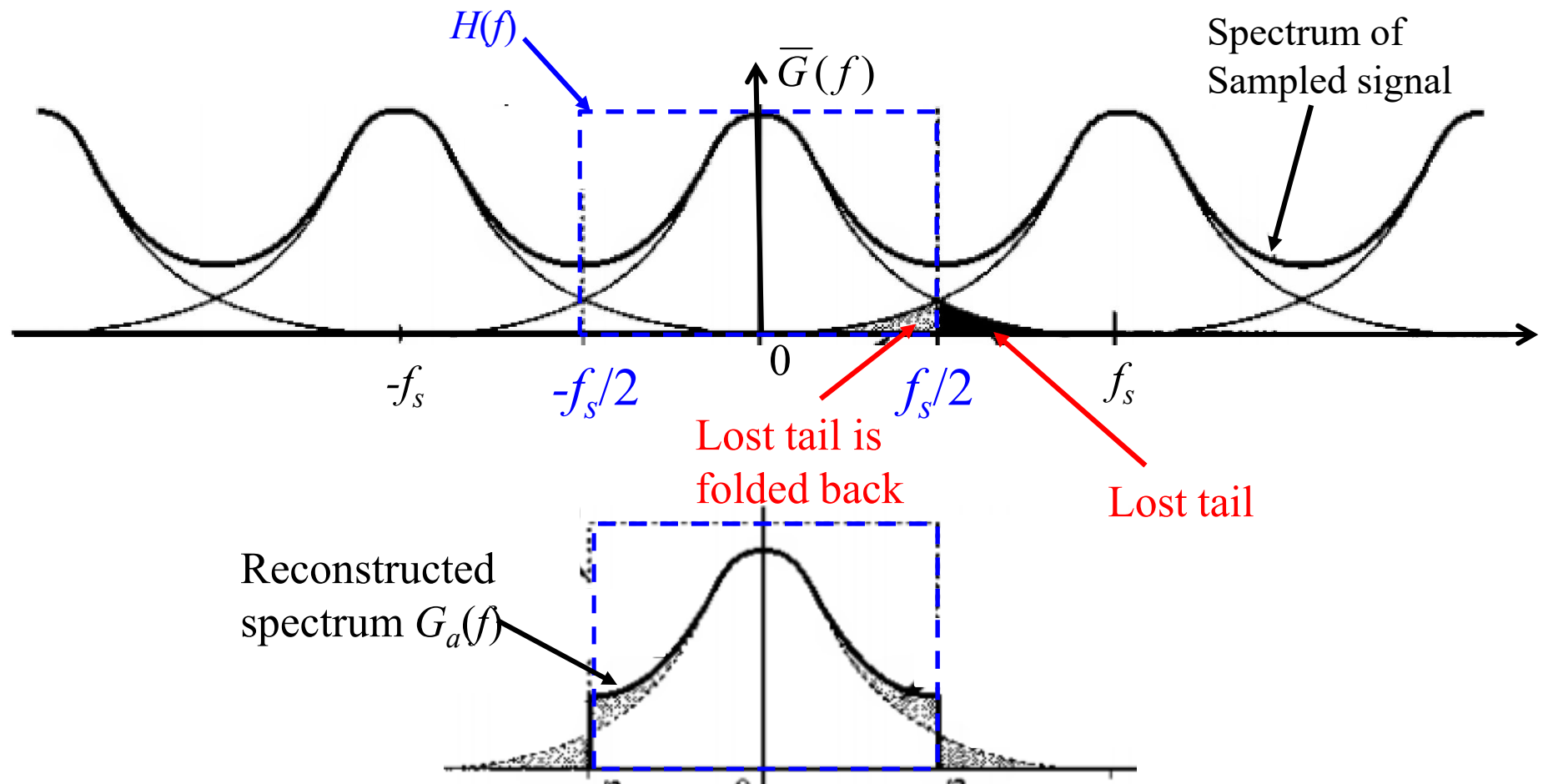


Ideal reconstruction filter CANNOT separate $G(f)$, rather generates $G_a(f)$ which spectrum of **aliased signal** $g_a(t)$

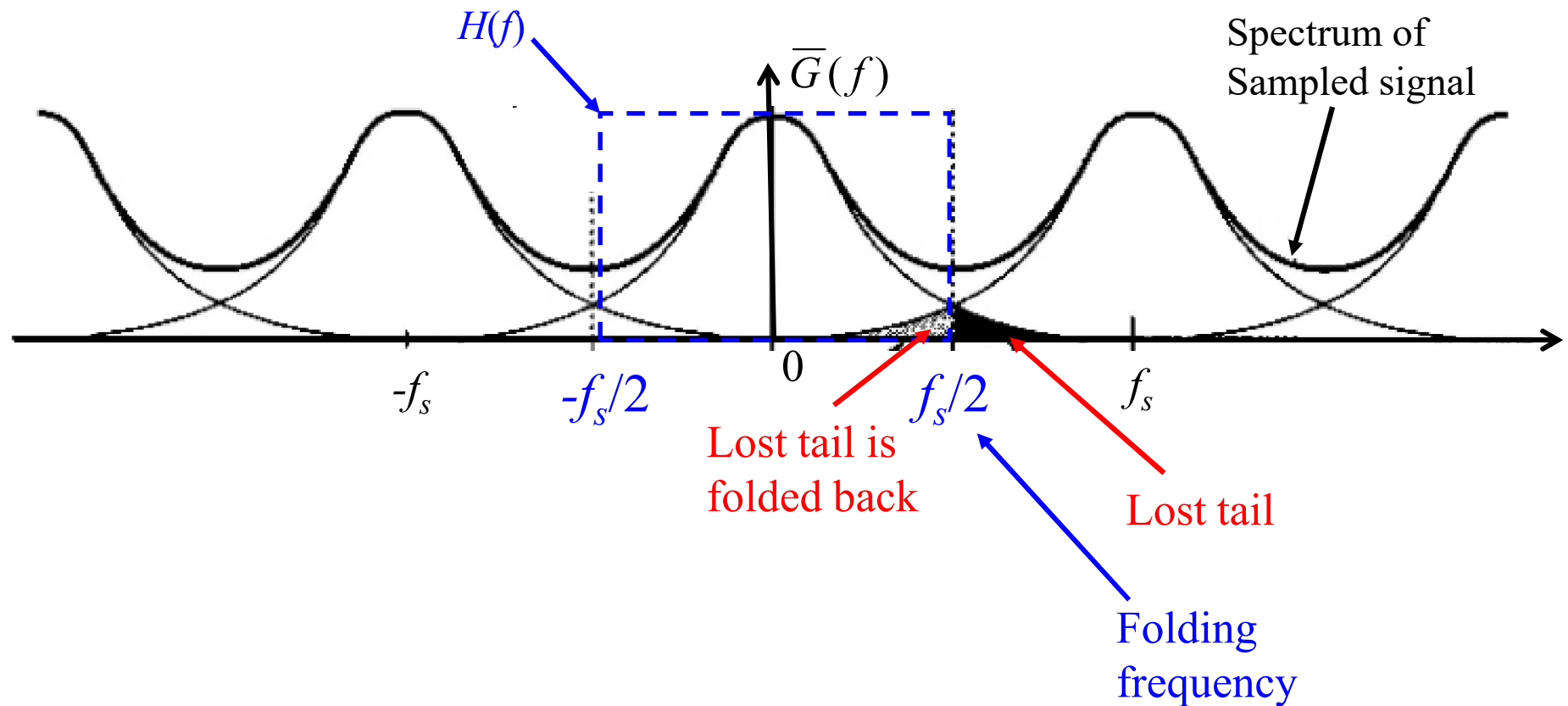
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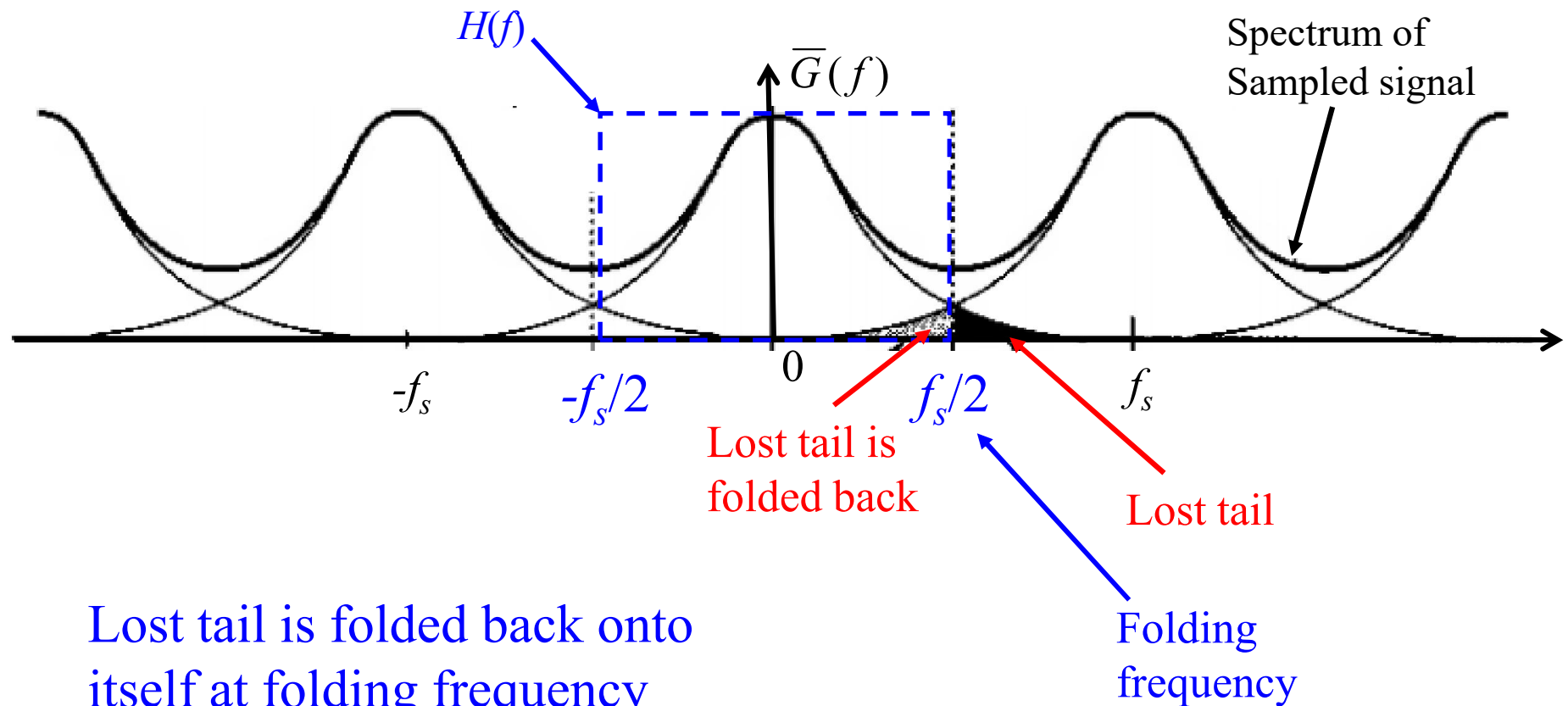
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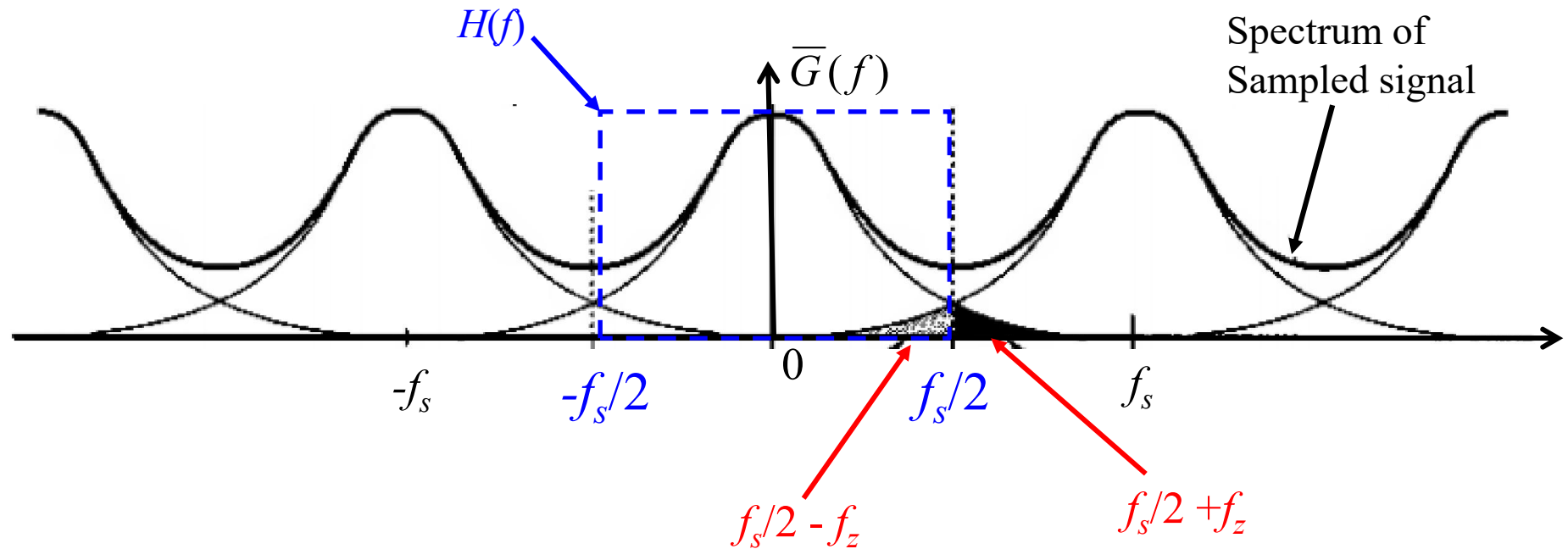
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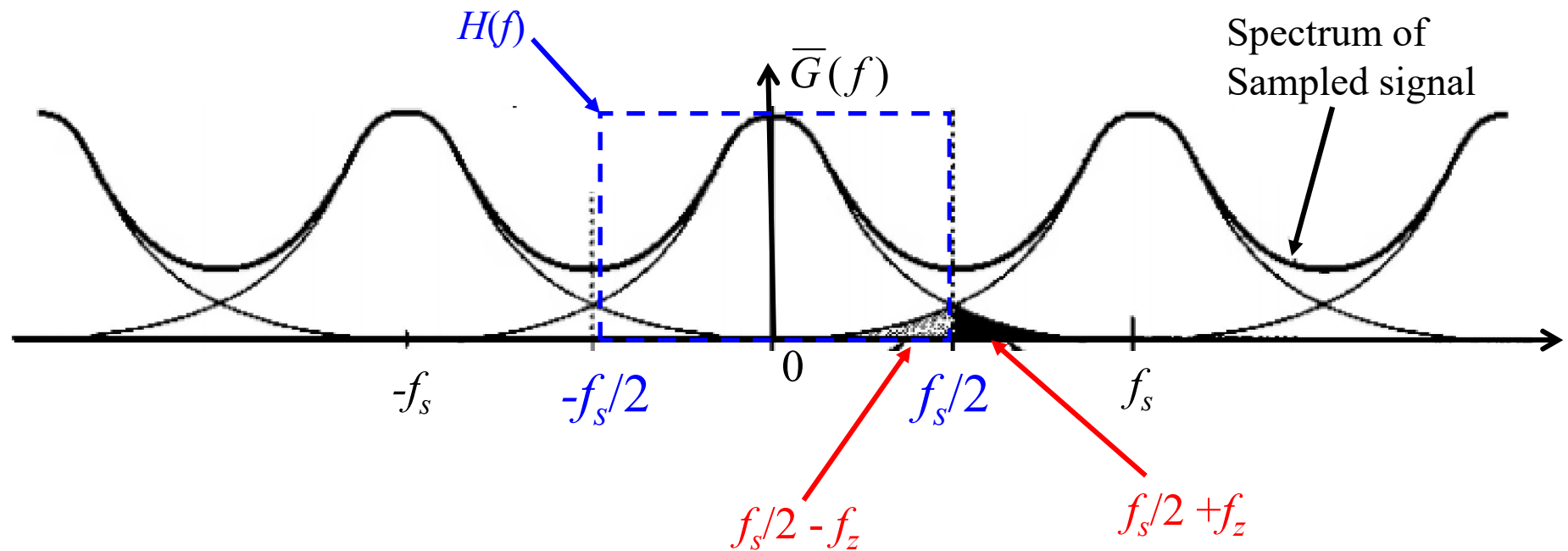


Wrong Assumption on Bandwidth: Treachery of Aliasing



Component at $f_s/2 + f_z$ is folded
back to $f_s/2 - f_z$

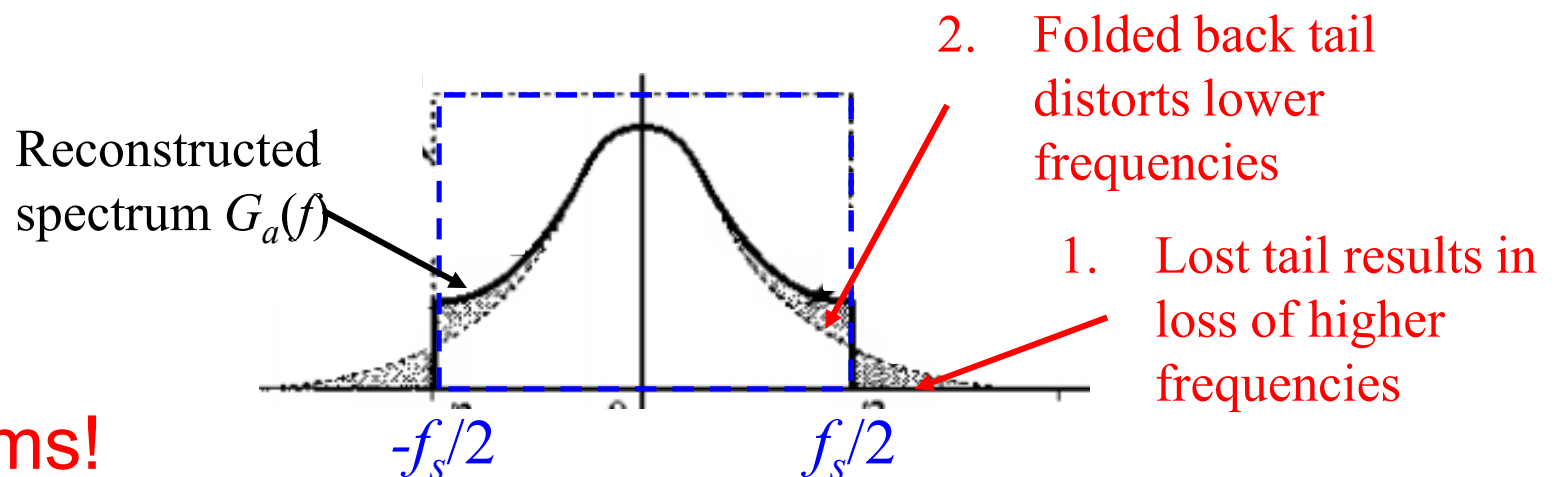
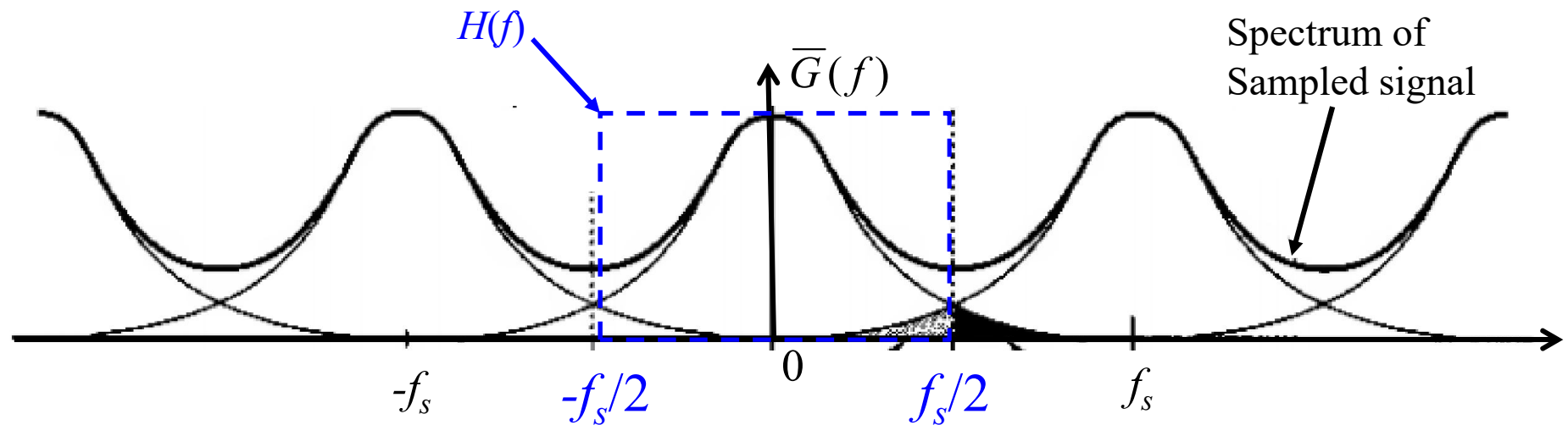
Wrong Assumption on Bandwidth: Treachery of Aliasing



Component at $f_s/2 + f_z$ is folded
back to $f_s/2 - f_z$

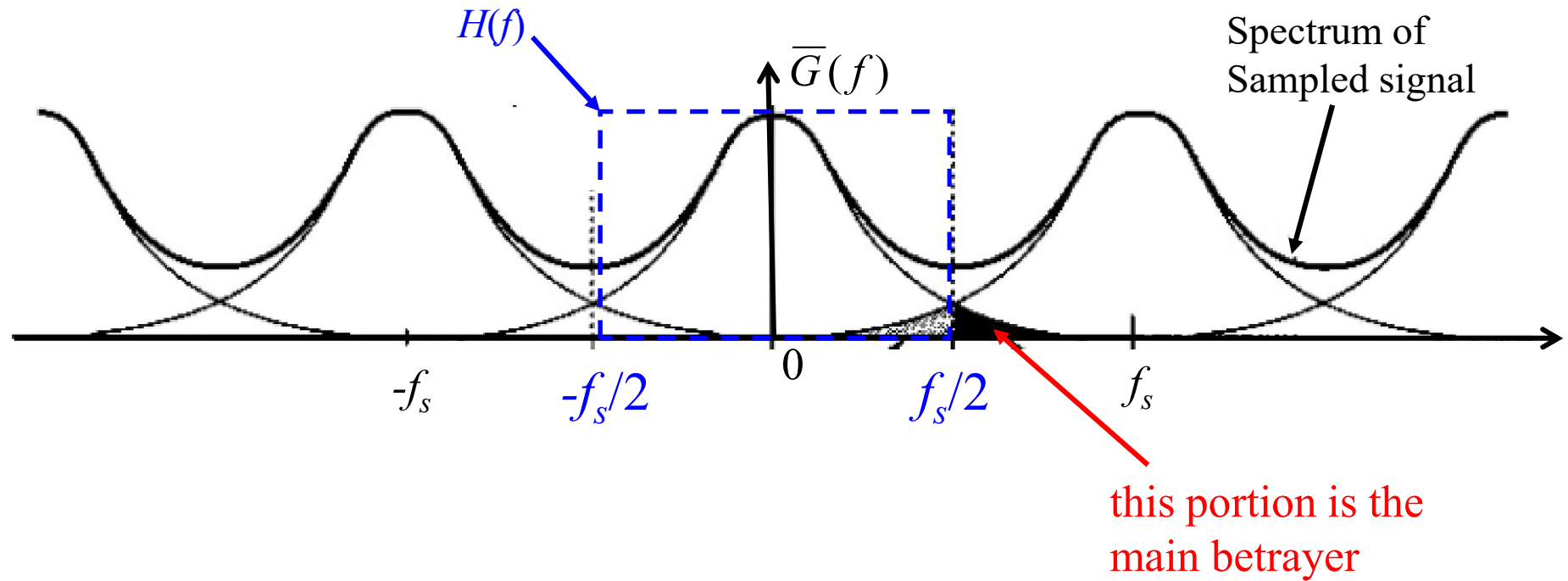
Tail folding is known as
spectral folding or aliasing

Wrong Assumption on Bandwidth: Treachery of Aliasing

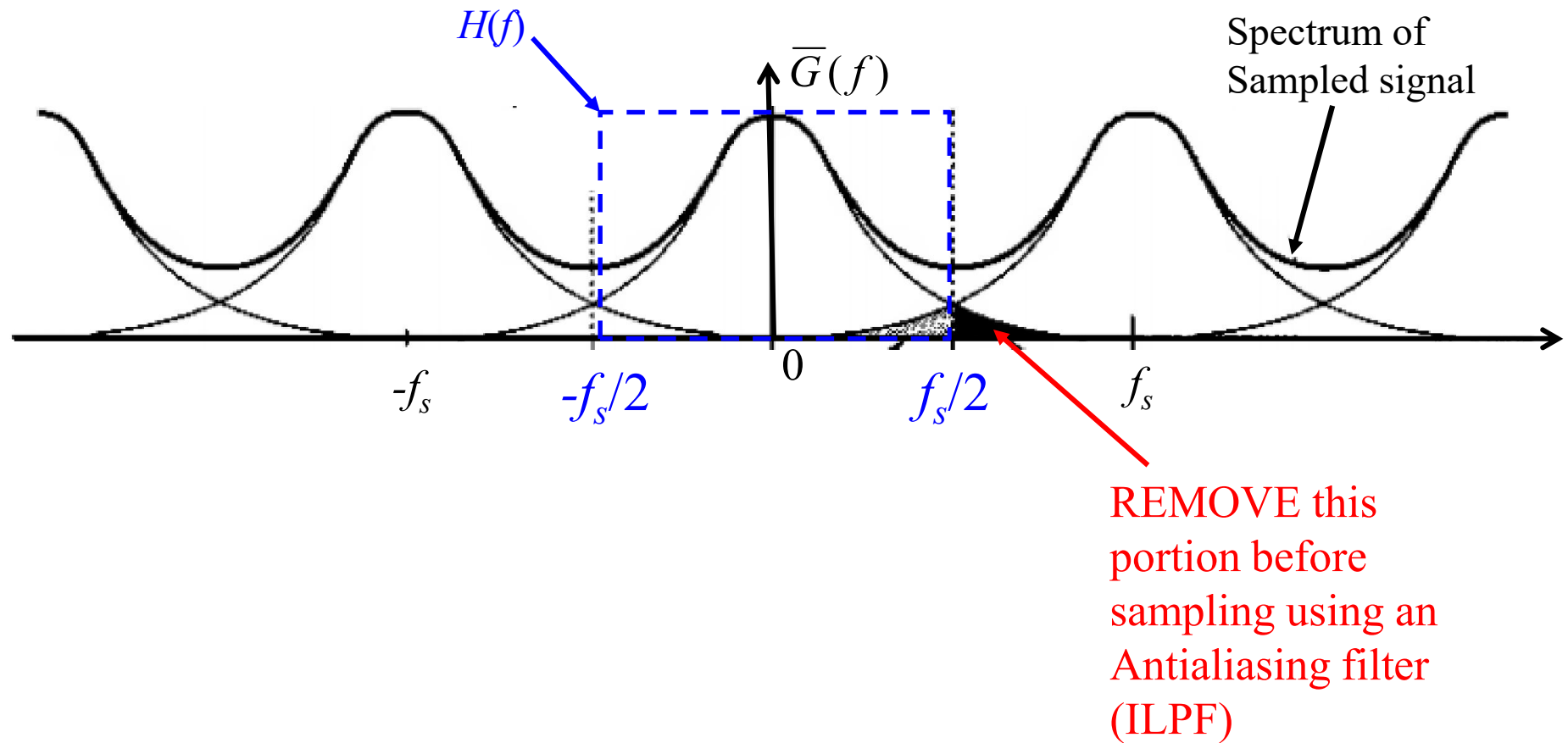


**TWO
Problems!**

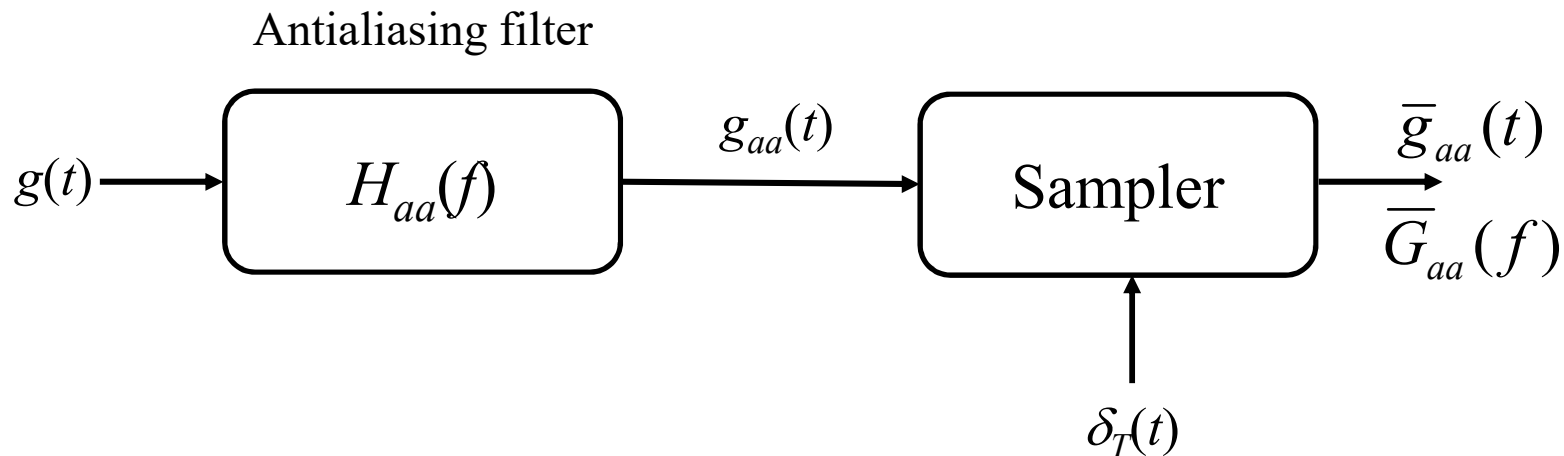
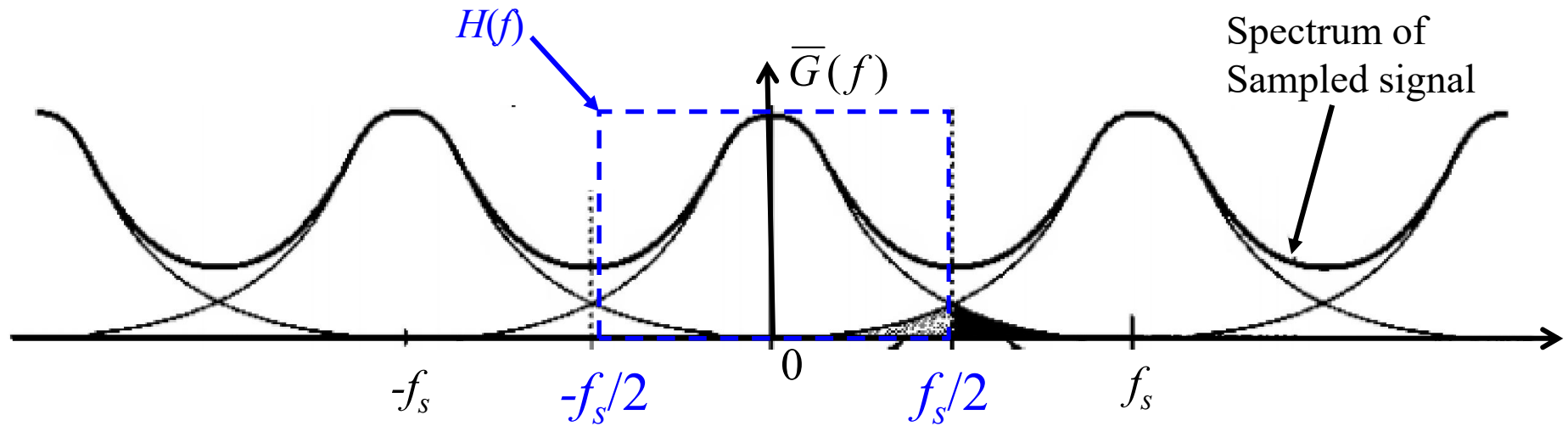
Combating Aliasing



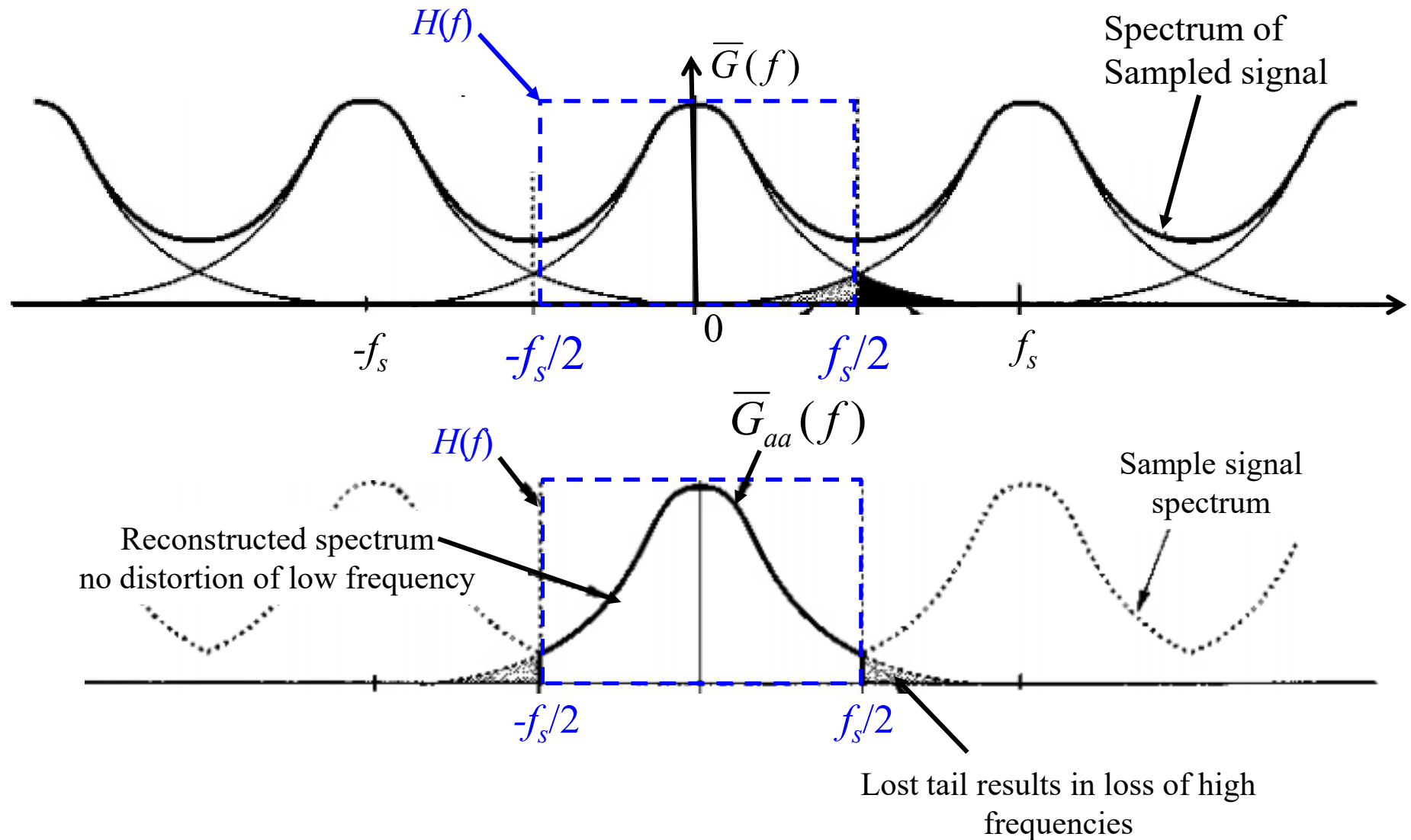
Combating Aliasing



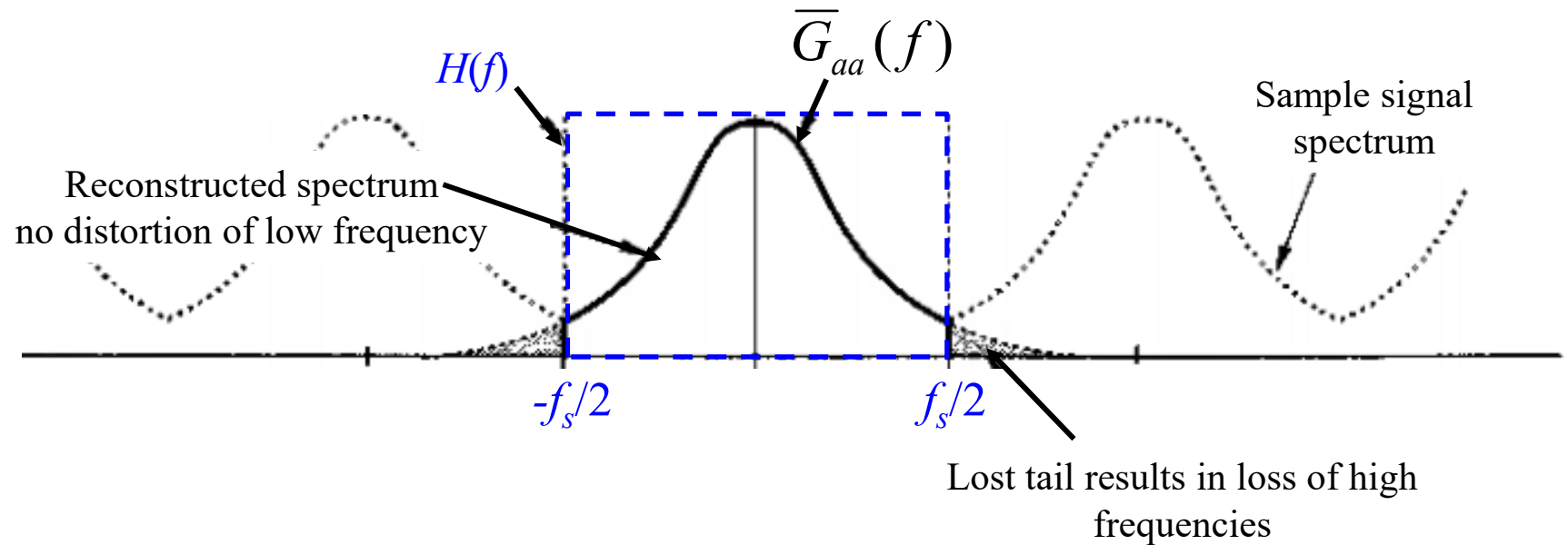
Combating Aliasing



Combating Aliasing



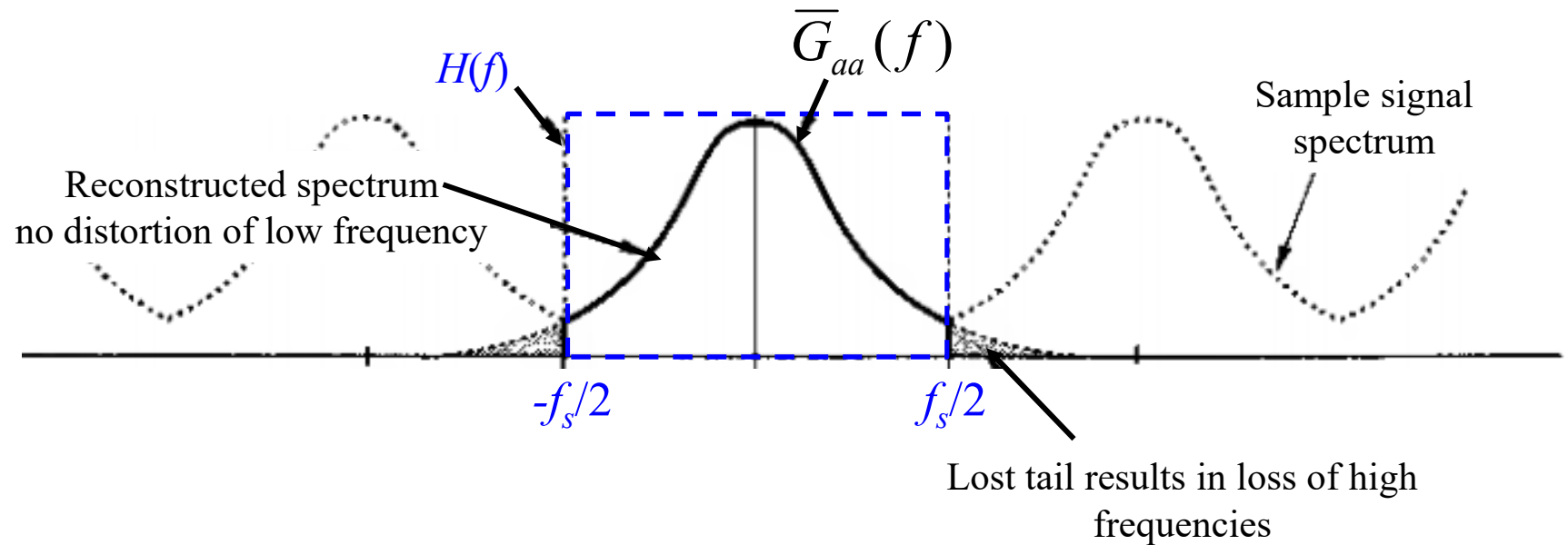
Combating Aliasing



$$\overline{G}_{aa}(f) = G(f) \quad |f| < f_s / 2$$

Advantage: suppressed
component cannot reappear
in low frequency
components

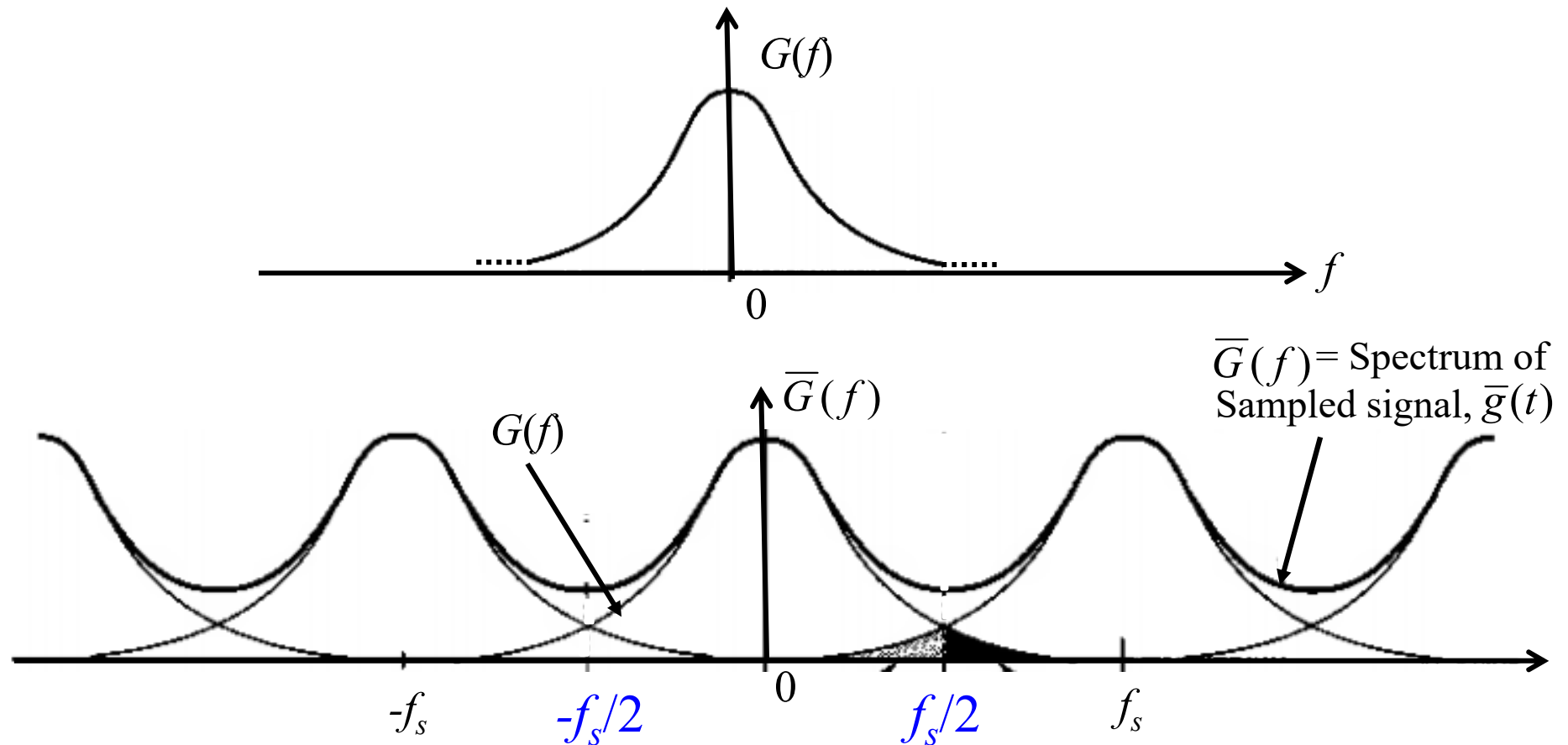
Combating Aliasing



Other characteristics:

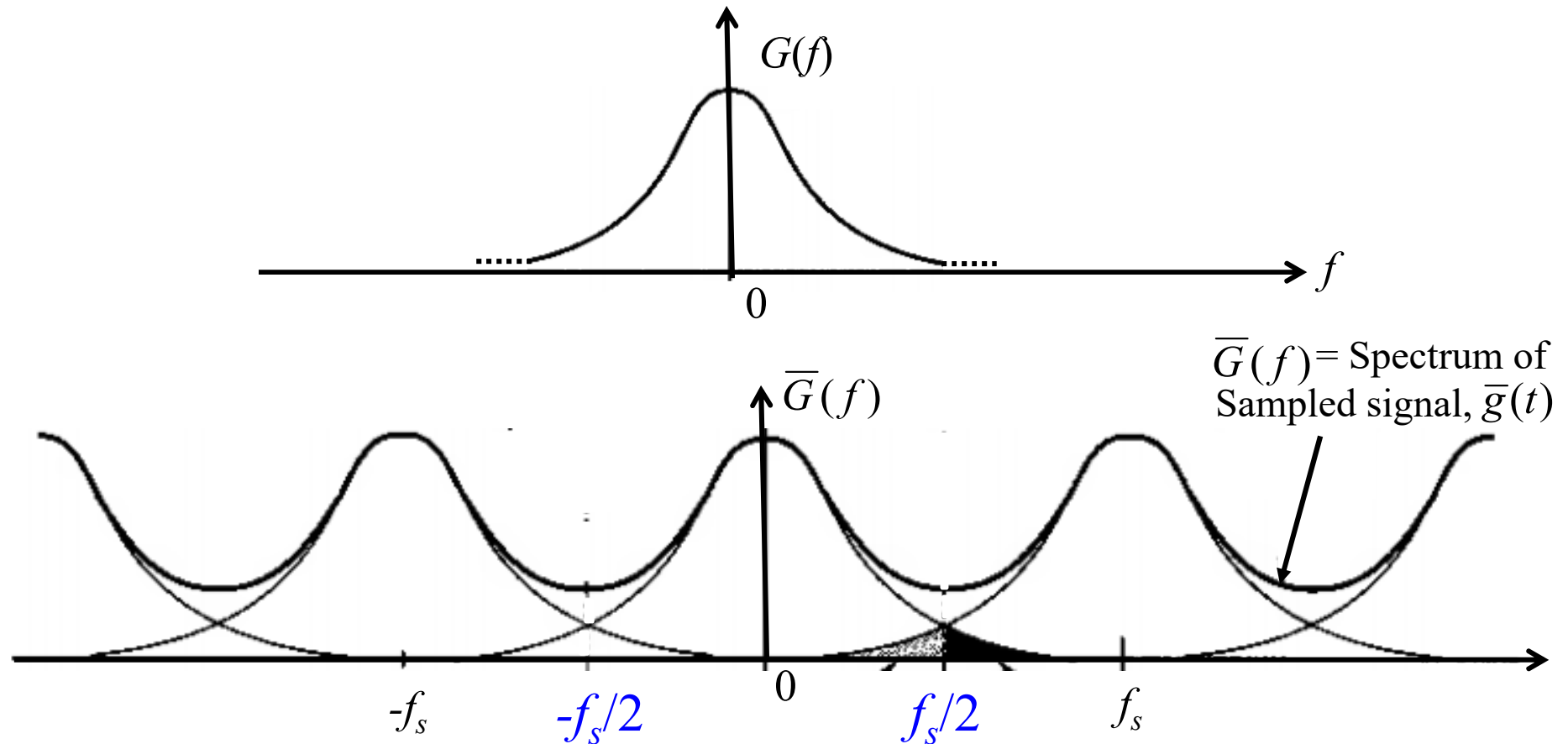
- Antialiasing filter band-limits $g(t)$ to $f_s/2$
- Reduces noise too, otherwise noise might be folded back to low frequency components
- Noises beyond $f_s/2$ are suppressed

Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

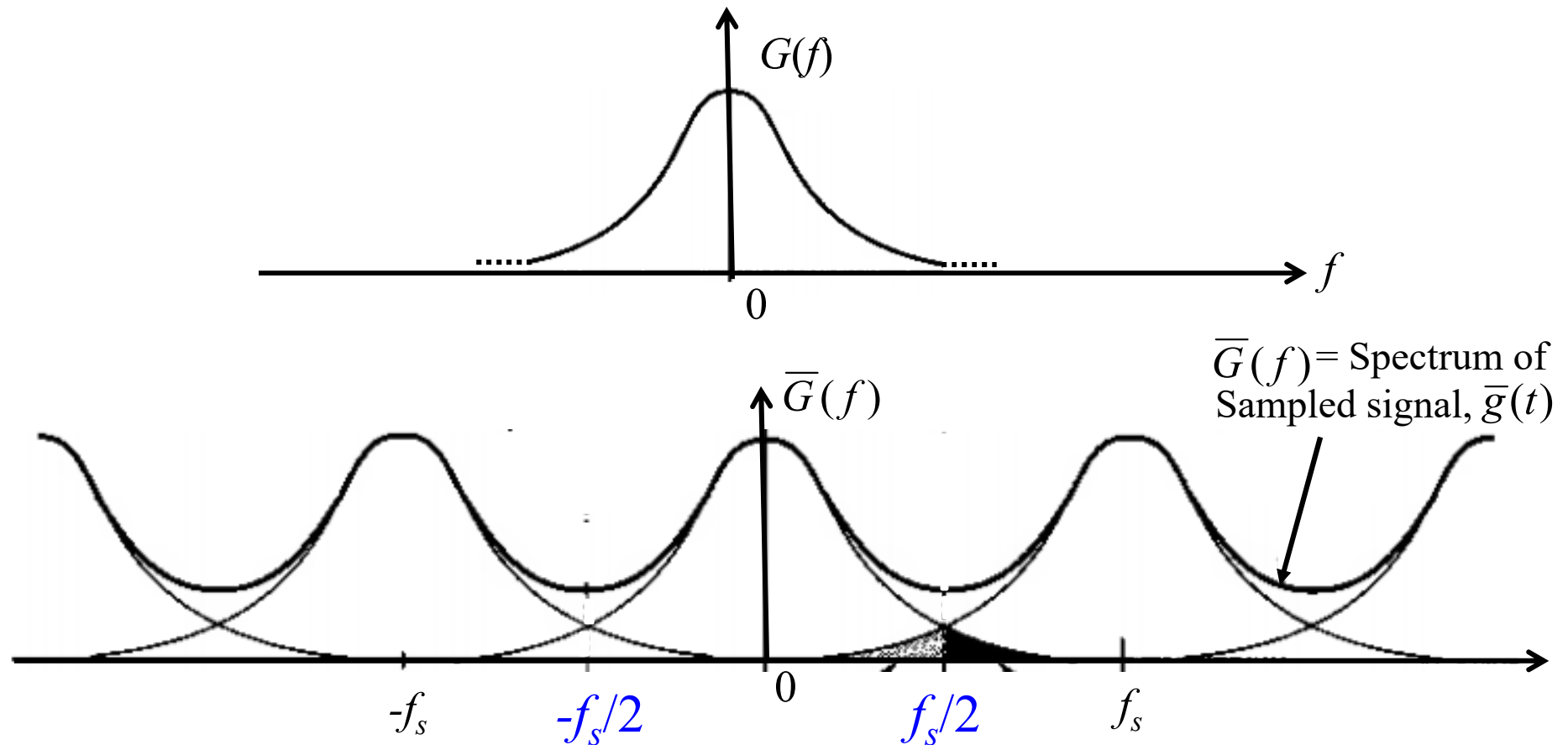
Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

$f_s \ll$ Nyquist rate

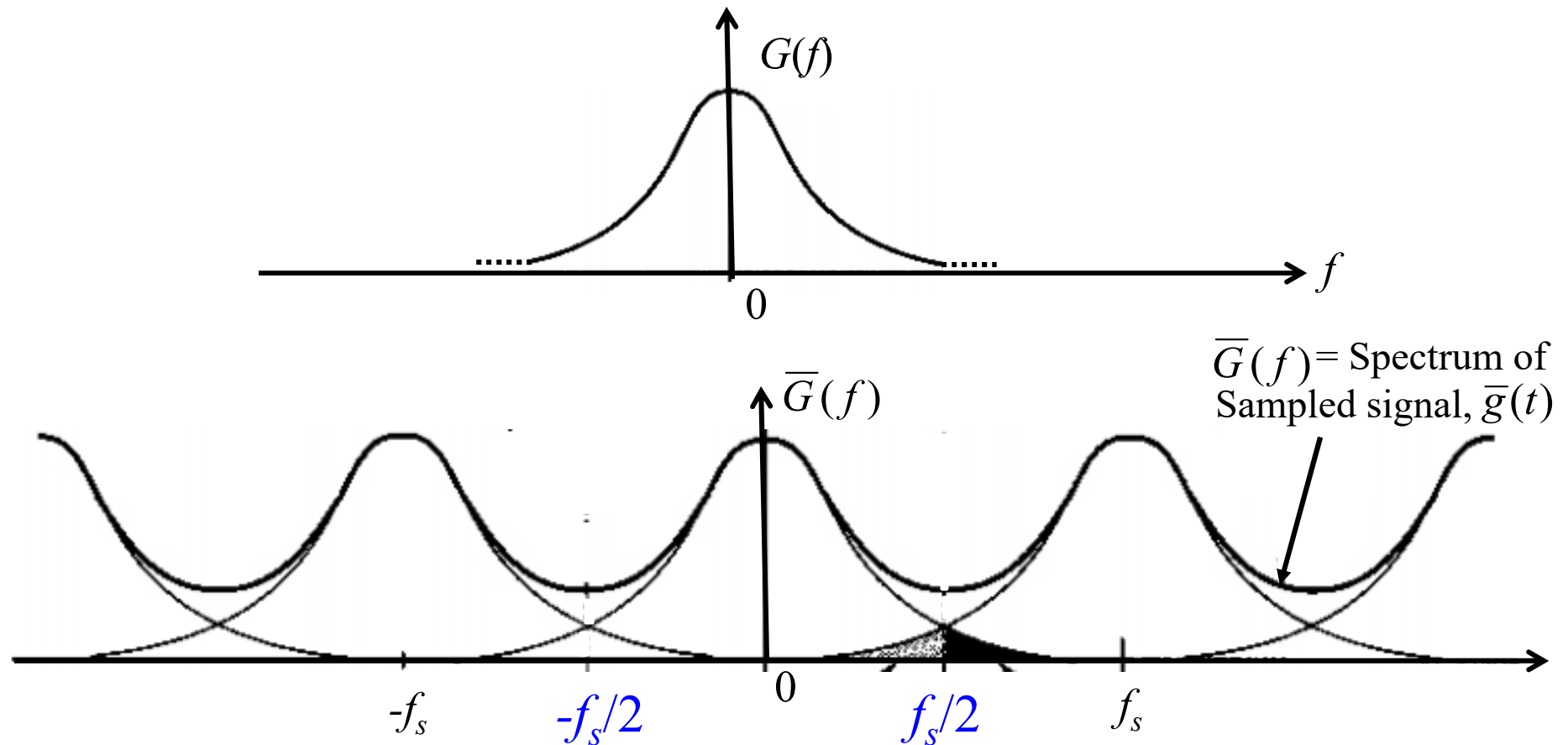
Sampling Effect: Non-band-limited signal to band-limited signal



Bandwidth of $G(f)$ is practically infinite

$\bar{G}(f)$, that means, spectrum of $\bar{g}(t)$ consists of overlapping cycles of $G(f)$

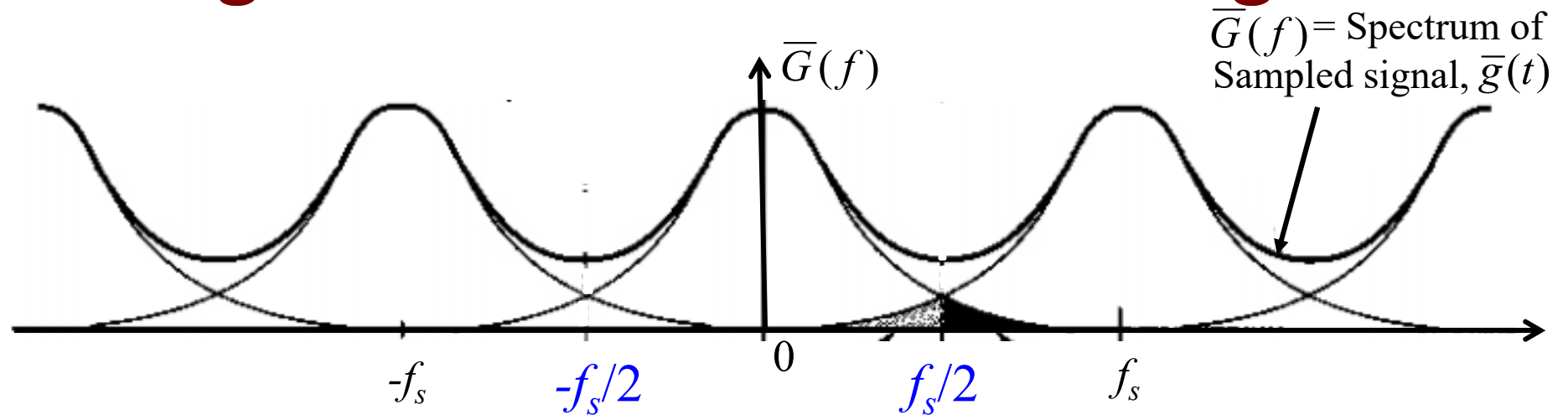
Sampling Effect: Non-band-limited signal to band-limited signal



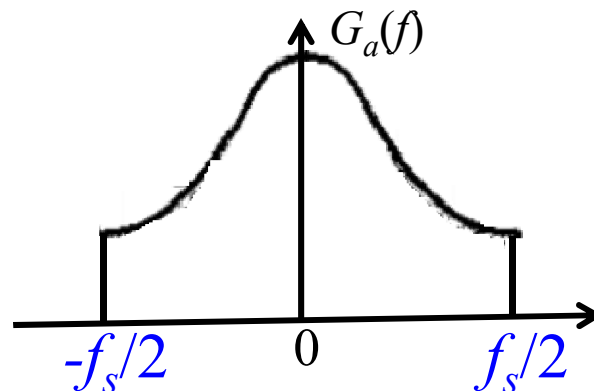
Bandwidth of $G(f)$ is practically infinite

$\bar{g}(t)$ are sub-Nyquist samples of $g(t)$!

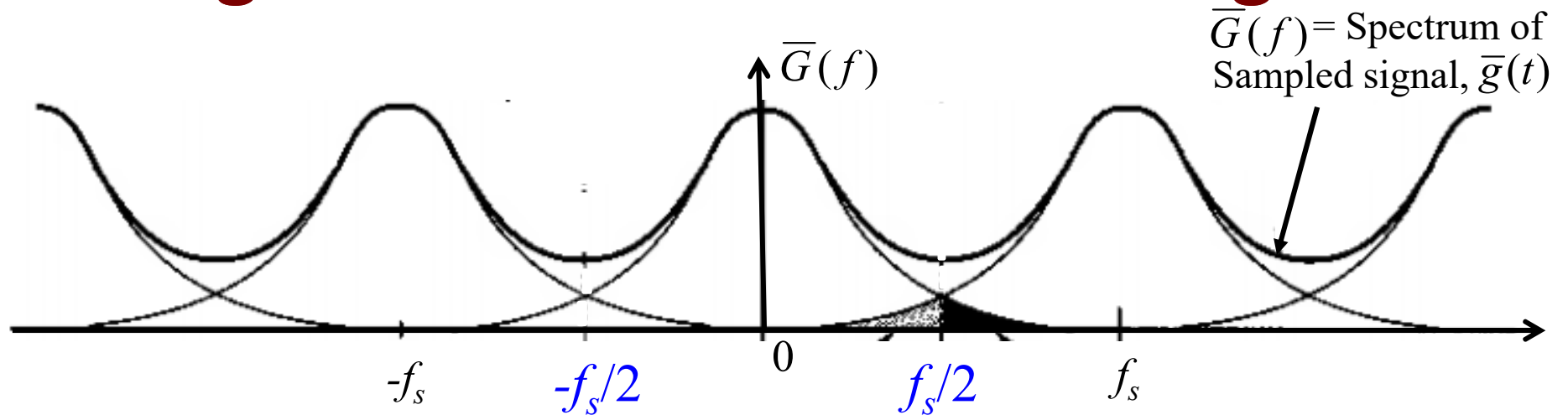
Sampling Effect: Non-band-limited signal to band-limited signal



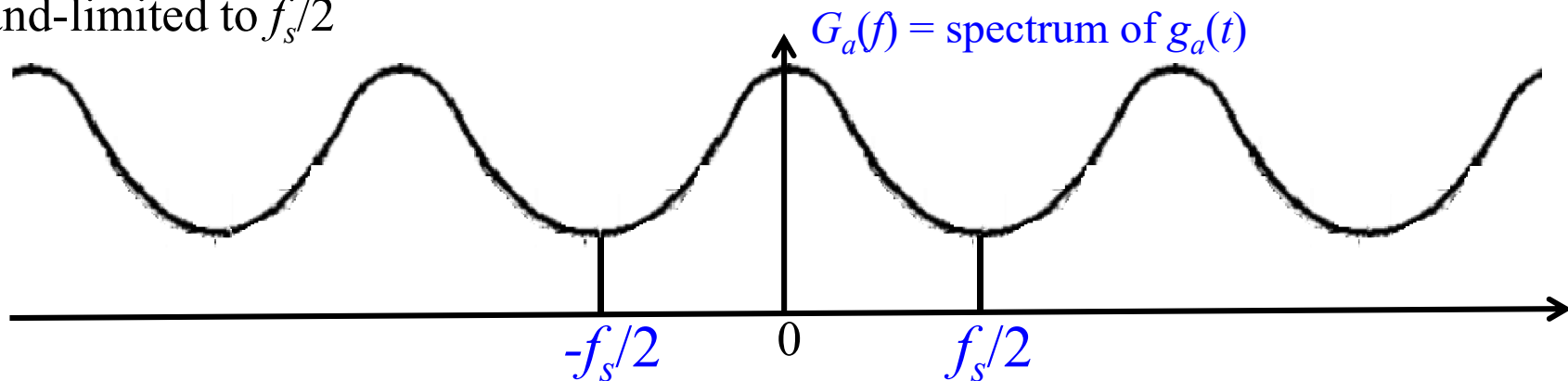
The **above bold line spectrum** can be viewed as **repetition** of the following $G_a(f)$ band-limited to $f_s/2$



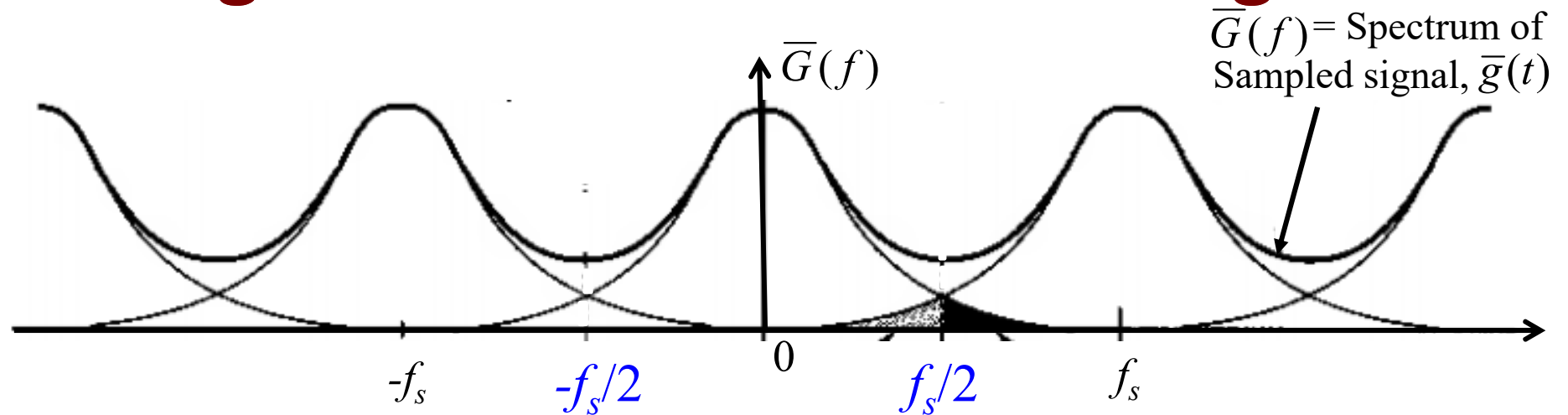
Sampling Effect: Non-band-limited signal to band-limited signal



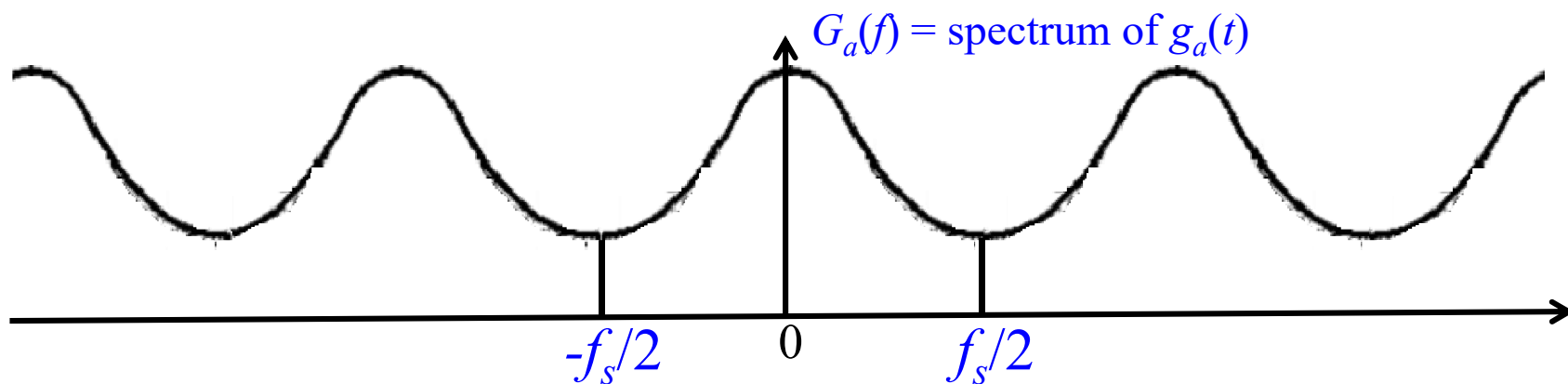
The above **bold line** spectrum can be viewed as repetition of the following $G_a(f)$ band-limited to $f_s/2$



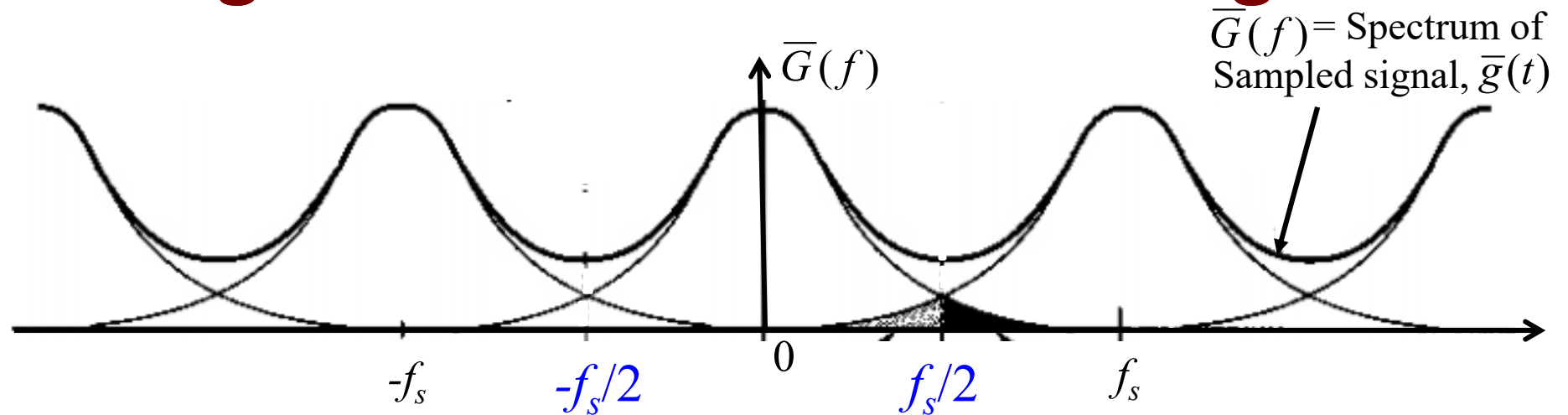
Sampling Effect: Non-band-limited signal to band-limited signal



$\bar{g}(t)$ are Nyquist samples of $g_a(t)$



Sampling Effect: Non-band-limited signal to band-limited signal



Sampling a non-band-limited signal $g(t)$ at f_s is equivalent to Nyquist sampling of some signal $g_a(t)$ band-limited to $f_s/2$